



Genome-wide identification and expression profiling of the *Subtilisin* (*SBT*) gene family reveals their involvement under abiotic stress (drought and cold) in *Cicer arietinum* L

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Abstract

Subtilisin-like serine proteases (SBTs) play pivotal roles in plant development and stress responses by regulating proteolytic processes. In this study, a comprehensive genome-wide analysis of the *SBT* gene family was performed in *Cicer arietinum* (desi and kabuli cultivars). The goal was to investigate their physiochemical characteristics, evolutionary relationships, structural and functional analysis and expression patterns under abiotic stress (drought and cold). A total of 69 *CaDSBT* and 82 *CaKSBT* genes were identified in desi and kabuli cultivars, respectively. The SBT proteins showed variations in amino acid numbers, isoelectric point, and subcellular localization, though most were found to be stable, basic, and hydrophilic. Phylogenetic analysis categorized the *SBT* genes into six groups (Group 1–6) with Group 1 having maximum number of genes in both cultivars. Chromosomal mapping revealed an uneven distribution, with Chromosome 7 containing highest number of genes. Promoter analysis identified abundant stress- and hormone-responsive elements, particularly in response to abiotic stress (especially light, defense, and general stress) and phytohormones (salicylic acid, MeJA, abscisic acid, and auxin). Gene ontology analysis showed enrichment in biological processes related to proteolysis and cellular metabolism. Gene duplication analysis indicated that purifying selection was the primary evolutionary force for the *SBT* gene family in chickpea. Physiological assessments confirmed that stress exposure resulted in a reduction in relative water content (RWC), and an increased in malondialdehyde and electrolyte leakage, indicating enhanced oxidative stress. Expression profiling revealed that *CaDSBT59*, *CaKSBT47*, and *CaKSBT77*, were strongly upregulated under drought and cold stress. These findings provide valuable insights into the potential of *SBT* genes for improving stress tolerance and developing future chickpea improvement strategies under stressful conditions.

Keywords Cold stress, · *Cicer arietinum* · Gene expression · Subtilisin -like proteases

Abbreviation:

SBT Subtilisin-like proteases
qRT-PCR Quantitative real-time polymerase chain reaction
GRAVY Grand average of hydropathicity

Chr Chromosome
MEME Multiple em for motif elicitation
HSPs High-scoring pairs
GO Gene ontology

Introduction

Plant subtilases (SBTs) are a highly conserved family of serine proteases that act as pivotal regulators of diverse biological processes, including stress adaptation, growth, and protein maturation (Wang et al. 2023). They are also known to play crucial roles in plant development, such as seed germination, cell division, tissue differentiation, and senescence (Galotta et al. 2019; Paulus et al. 2020). Beyond

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their developmental functions, SBTs are critical mediators of abiotic stress responses and signal transduction (Paulus et al. 2020), with many *SBT* genes showing strongly modulation by abiotic stresses like cold and drought (Wang et al. 2023). During stress, SBT-mediated protein degradation is crucial for tolerance, as it supplies amino acids for energy and the de novo synthesis of stress-responsive proteins (Dai et al. 2023).

SBT proteinases, which are frequently expressed under stress conditions, are crucial for autophagy processes. Their involvement in responses to environmental stimuli, particularly abiotic stress, underscores their importance (Paulus et al. 2020). With the escalating threat of abiotic stresses to global crop productivity under climate change, the functional significance of *SBT* genes has been characterized across numerous plant species. The *SBT* gene family has been well documented in *Arabidopsis thaliana*, *Zea mays*, *Gossypium hirsutum* (Zhao et al. 2023; Dai et al. 2023; Hou et al. 2023). For example, in *Arabidopsis thaliana*, *AtSBT1.2* regulates stomatal density, which is crucial for drought adaptation (Zhao et al. 2023), while overexpression of *AtSBT3.3* increases tolerance to environmental stresses (Ramírez et al. 2013). In *Zea mays*, several *SBTs* (*ZmSBT17*, *ZmSBT18*, and *ZmSBT41*) are downregulated by cold stress, and others (*mSBT7*, *ZmSBT8*, *ZmSBT18*, *ZmSBT35*, and *ZmSBT48*) are downregulated by drought suggesting their involvement in cold acclimation (Hou et al. 2023).

SBT genes have also been well-documented in legumes, including *Medicago truncatula*, *Pisum sativum*, *Glycine max*, *Arachis hypogaea*, *Phaseolus vulgaris* (Li et al. 2024; He et al. 2024; Budić et al. 2013). In soybeans, *SBT* genes play critical regulatory roles in signaling pathways associated with abiotic stress tolerance, making them key targets for genetic improvement under adverse conditions (He et al. 2024). In *Arachis hypogaea*, several *SBTs* (*SBT1 AhSBT39*, *AhSBT53*, *AhSBT92*, and *AhSBT115*) are upregulated by drought and cold, indicating their direct involvement in stress responses (Li et al. 2024). Furthermore, in *Vigna radiata* *SBT* genes are significantly upregulated under both high and low temperatures, which correlates with protein turnover during abiotic stress (Olivares et al. 2021). *Subtilases* from *Phaseolus vulgaris*, such as *PvSLP2*, increase chymotrypsin activity during dehydration, implicating them in drought-induced senescence and proteolytic adjustments (Budić et al. 2013). This suggests a regulatory role for *SBT* in modulating plant longevity and stress survival under water deficit.

Chickpea (*Cicer arietinum* L.), a vital legume, that is highly susceptible to abiotic stresses like drought, cold, and salinity. These stresses are responsible for substantial global yield reductions, with drought and salinity alone causing losses of up to 60% (Sistu et al. 2024). Among

these, drought is a particularly severe threat, leading to inhibited plant growth and reduced crop quality and seed yield. Drought stress is known to induce biochemical and physiological alterations in chickpea, including reduced growth rates, decreased chlorophyll content, and lowered ascorbic acid levels. (Asati et al. 2024). Key physiological parameters such as relative water content (RWC), membrane stability index (MSI), chlorophyll content, and leaf area index (LAI) are used to elucidate yield variability under stress conditions.

For comprehensive understand the *subtilase* (*SBT*) gene family in chickpea, we performed genome-wide identification and characterization across the desi and kabuli cultivars genome. This study provides a novel, in-depth analysis of the sequence characteristics, phylogenetic relationships, gene structure, promoter elements and conserved motifs of all identified *SBT* genes in both chickpea varieties. Additionally, we used Gene ontology (GO) to determine their roles in various biological processes and molecular functions. Most importantly, this research uniquely examines the regulatory roles of these genes under specific stress conditions, as we evaluated the expression levels of selected *SBT* genes in leaf samples from both desi and kabuli chickpea cultivars subjected to varying concentrations of drought stress (10, 20, 30 $\mu\text{mol/L}$) and cold stress (4 $^{\circ}\text{C}$). This study offers significant insights into the chickpea *SBT* gene family, revealing their potential for enhancing drought and cold tolerance in chickpea improvement strategies.

Materials and methods

Genome-wide identification of *SBT* gene family in the genome of *Cicer arietinum*

Genome-wide identification of *SBT* gene family in the genome of *Cicer arietinum*

To identify *subtilase* (*SBT*) genes in chickpea genome, the public reference genomes of both desi (ICC 4958, v2.0) and kabuli cultivars (CDC Frontier genome assembly GCF_000331145.1) were downloaded from the Chickpea database (<http://www.cicer.info/databases.php>). The reason for selection of two contrasting cultivars of chickpea for present work is that both have well documented phenotypic and physiological differences, including drought resistance. A subsequent BLASTp search was performed using previously identified *SBTs* protein sequences as queries from plants of *Arabidopsis thaliana* (*AtSBT*) and *Arachis hypogaea* (*AhSBT*). These were selected based upon annotation quality, functional complementarity and phylogenetic proximity of *Arabidopsis thaliana* and *Arachis hypogaea*,

respectively with the genome of *Cicer*. The genome versions of *Arabidopsis thaliana* and *Arachis hypogaea* used for annotation are TAIR10 and Tifrunner v2.0, respectively. Initial putative SBT-encoding genes in chickpea were identified using an e-value cut-off of 10^{-5} (as a strategic balance) to select high-scoring pairs (HSPs) including those that have diversified significantly. The final selection was refined by retaining HSPs with bit scores greater than 100, and all duplicate entries were removed to ensure a non-redundant dataset for downstream analysis. All identified *SBT* genes were then analyzed for conserved domains using both the Pfam database and NCBI's Conserved Domain Search database (<https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>). Transmembrane helices of the predicted SBT protein sequences were computationally analyzed using the online tool SOSUI/G ver.1.1 (<https://harrier.nagahama-i-bio.ac.jp/sosui/sosuiG/sosuiGsubmit.html>). Subsequently, the genome-wide distribution of the *SBT* genes was extracted and visualized with TBtools-II v2.210 software (Chen et al. 2020).

Characterization of SBT proteins in the genome of *Cicer arietinum*

To examine the characteristic chickpea SBT proteins, their amino acid number, relative molecular weight (MW), isoelectric points (pI), grand average of hydropathy (GRAVY), and instability index were calculated using the ExPASy version 3.0 database (https://web.expasy.org/compute_pi/). Additionally, the subcellular localization of these SBT proteins was predicted with the WoLFPSORT:v0.2 tool (http://www.genscript.com/psort/wolf_psport.html).

Phylogenetic analysis of the SBT encoding genes

To examine the evolutionary relationships of chickpea subtilase (SBT) proteins, protein sequences from desi (CaDSBTs) and kabuli (CaKSBTs) chickpea cultivars were aligned with previously characterized SBT proteins from *Arabidopsis thaliana* (AtSBTs) and *Arachis hypogaea* (AhSBT) using the ClustalW program (Thompson, 1994). Subsequently, a phylogenetic tree was constructed in MEGA software version 11.0.14 (Tamura et al. 2021). The phylogenetic analysis employed 1000 bootstrap replicates to assess tree robustness. The final, filtered genes were numerically labelled for clarity.

Chromosomal location, gene structure, and conserved motif analysis of SBT encoding genes in *C. arietinum*

Using gene coordinates extracted from the General Feature Format version 3 (GFF3) file, TBtools software (<https://github.com/CJ-Chen/Tbtools>) was employed to determine the chromosomal localization, motif patterns, and intron–exon structure of the identified SBT-encoding genes in chickpea. Subsequently, conserved motifs within these SBT proteins were discovered through in silico analysis using the Multiple Em for Motif Elicitation (MEME motif) search tool version 5.5.8 (<http://mem-esuite.org/tools/meme>) (Bailey et al. 2009). The parameters for motif identification were set to a maximum of 10 motifs, with any number of repetitions, and motif widths ranging from a minimum of 6 to a maximum of 50 base pairs.

Gene ontology analysis of SBT encoding genes

Functional analysis of chickpea protein sequences (from both desi and kabuli cultivars) was performed using the Blast2GO bioinformatics platform (<http://www.blast2go.com>). This involved loading the amino acid sequences into the program and conducting Gene Ontology (GO) analysis. The resulting GO classifications were examined across the three major categories such as biological processes, molecular functions, and cellular components, to infer the prospective roles of these proteins.

Cis-regulatory elements analysis of SBT encoding genes

To elucidate the promoter structure of *SBT* genes in *Cicer arietinum*, a 2000 bp upstream region from the translation start site of each gene was extracted using TBtools v2.2210 (Chen et al. 2020). These promoter sequences were analyzed for cis-regulatory elements using the PlantCARE database. Subsequently, the identified cis-elements were categorized based on their association with developmental processes, phytohormone signaling, and abiotic stress responses.

Ks/Ka-based analysis of homologous gene pairs of *SBT* genes

Putative homologous gene pairs of *SBT* genes between desi and kabuli chickpea were identified manually based on the constructed phylogenetic tree and alignment was performed using ClustalW program in MEGA software version 11. Subsequently, the non-synonymous/synonymous (Ka/Ks) substitution rate ratio for these gene duplication pairs was calculated using TBtools software (Chen et al. 2020), with

corresponding protein and genomic sequences as input. The approximate divergence times of these duplication events were then estimated using the formula $T = Ks / (2\lambda)$, where λ (the clock-like rate of synonymous substitution for chickpea) is 6.1×10^{-9} substitutions/site/year.

Structural analysis of *C. arietinum* SBT proteins

To examine the three-dimensional structures of chickpea SBT proteins, homology modelling was conducted using the Phyre2 web server (<http://www.sbg.bio.ic.ac.uk/phyre2>). The amino acid sequences of SBT proteins from both desi and kabuli chickpea cultivars were submitted to the Phyre2 platform for determination of their tertiary structures (Kelley et al. 2015). The quality of the resulting models was assessed based on several metrics, including confidence scores, alignment coverage, structural consistency (e.g., alpha-helix and beta-strand content), and stability levels.

Drought and cold stress responses analysis in *C. arietinum*

a) Plant materials and stress treatments

To investigate stress responses in chickpea, seeds of the desi cultivar ICC4958 and kabuli cultivar L550 were cultivated in pots containing vermiculite under controlled conditions with three biological replicates per treatment. On the seventh day of growth, plants were subjected to drought and cold stress. Drought stress was induced by utilizing polyethylene glycol (PEG 6000) at 10%, 20%, and 30% (w/v) concentrations, while cold stress was induced by placing plants at 4 °C. After 24 h of treatment, leaf samples were harvested for subsequent physiological and gene expression analyses. A 24-h exposure to cold stress is a common choice for an experiment because it captures the initial, or primary, responses of the plant. This timeframe is sufficient to observe the immediate changes in gene expression and metabolism as the plant first reacts to the stress.

b) Determination of relative water content (RWC)

The relative water content (RWC) of chickpea leaf samples was determined using method described by Yadav et al. (2022). Briefly, after leaf sample collection, the fresh weight (FW) of the tissue was recorded. Subsequently, samples were rehydrated in deionized water for 24 h, surface-dried, and their turgid weight (TW) measured. Further, samples were oven-dried at 80 °C for 24 h to constant weight, and their dry weight (DW) was recorded. All measurements were performed in triplicates. The RWC was calculated using the following formula:

$$RWC (\%) = (TW - DW) / (FW - DW) \times 100$$

c) Determination of electrolyte leakage index (ELI)

Membrane damage was quantified by measuring electrolyte leakage (EL), following the methodology described by Yadav et al. (2022). Briefly, 50 mg of fresh leaf tissues were submerged in 20 mL of deionized water and incubated overnight. The initial electrical conductivity (C1) of the medium was then measured using a conductivity meter. Subsequently, the samples were incubated at 100 °C for one hour to induce complete membrane disruption and release all electrolytes. The final electrical conductivity (C2) of the medium was then measured. The percentage of electrolyte leakage was calculated using the following formula:

$$\text{Electrolyte leakage index } (\%) = C1 / C2 \times 100$$

d) Determination of MDA content

Lipid peroxidation in chickpea leaf tissue is quantified by measuring malondialdehyde (MDA) content. Briefly, the analysis involved homogenizing 200 mg of fresh leaf samples in 5 mL of a solution containing 0.6% thiobarbituric acid (TBA) in 10% trichloroacetic acid (TCA). The homogenate is then heated at 100 °C for 15 min, followed by rapid cooling on ice. After cooling, the samples were centrifuged at 5000 rpm for 10 min. The supernatant's absorbance was measured spectrophotometrically at 450 nm, 532 nm, and 600 nm, using the TBA-TCA solution as a blank. The MDA concentration ($\mu\text{mol g}^{-1}$ fresh weight) was calculated using the formula:

$$MDA = 6.45 \times (OD_{532} - OD_{600}) - 0.56 \times OD_{450} \times Z$$

Gene expression analysis of SBT encoding genes in *Cicer arietinum*

Total RNA was extracted from leaf samples of both control and stressed chickpea plantlets using TRI Reagent (Sigma Life Science, St. Louis, MO), with three biological replicates per sample. The concentration and purity of the isolated RNA were assessed using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, Wilmington, DE). For gene expression analysis, six *SBT* (*subtilisin-like protease*) genes, one from each characterized group within the desi and kabuli chickpea genomes, were selected. We selected at least one representative gene from each major phylogenetic group and primary selection criteria for these gene was presence of presence of stress-related cis-elements to analyse the expression analysis with the highest potential

to play a significant role in stress tolerance in chickpea. Specific primers for each target gene were designed using Primer3 software (version 0.4), and their quality was validated with the OligoCalc tool (Table 1). Gene expression analysis was performed using a Real-Time PCR System (Applied Biosystems). The default settings for all reactions were 2 min at 50 °C, 10 min at 95 °C, 40 cycles of 15 s at 95 °C, and 1 min at 60 °C. The eukaryotic translation *elongation factor 1-alpha (elf-1)* gene was utilized as an endogenous control for normalization, as previously established by Garg et al. 2010. Relative fold changes in gene expression were calculated using the $2^{-\Delta\Delta CT}$ method. Graphical representations were generated to visualize the gene expression levels across different chickpea samples.

Results

Genome wide identification of *SBT* gene family reveals diversity between kabuli and desi chickpea cultivars

A total of 56 *AtSBT* genes from *Arabidopsis thaliana* and 122 *AhSBT* genes from *Arachis hypogaea* were used as query sequences in a BLASTp (v2.13.0) search to identify and curate SBT genes in the chickpea genome (desi and

kabuli cultivars). The analysis identified 69 *SBT* genes in desi chickpea (designated *CaDSBT1* to *CaDSBT69*) and 82 *SBT* genes in kabuli chickpea (designated *CaKSBT1* to *CaKSBT82*). The identified SBT proteins in chickpea were classified into six groups based on conserved domain characteristics. In the desi cultivar, 69 *CaDSBT* proteins were distributed across Group 1 (20), Group 2 (10), Group 3 (3), Group 4 (20), Group 5 (6), and Group 6 (8). Similarly, the kabuli cultivar exhibited a comparable distribution, with 82 *CaKSBT* proteins assigned to Group 1 (24), Group 2 (11), Group 3 (2), Group 4 (21), Group 5 (18), and Group 6 (5).

The transmembrane domains of the curated SBT proteins were predicted using SOSUI software. In desi chickpea, 23 SBT proteins were identified as membrane protein with transmembrane helices, while 46 were soluble proteins (Supplementary Fig. 1a). Notably, *CaDSBT33* contained three transmembrane helices, whereas *CaDSBT6* and *CaDSBT56* each had two transmembrane helices. In kabuli chickpea, out of 82 SBT proteins, 27 were identified to be membrane proteins and 55 were soluble. Among them, *CaKSBT46* was found to contain twelve transmembrane helices, while *CaKSBT19*, *CaKSBT42*, *CaKSBT76*, and *CaKSBT78* each possessed two. The difference in the number of helices between *CaKSBT46* (12 helices) and other SBTs strongly suggests unique membrane trafficking or singling roles for *CaKSBT46*.

Physicochemical analysis and subcellular localization of SBT proteins reveals less significant differences among desi and kabuli chickpea cultivars

A comprehensive analysis was conducted to characterize the physicochemical properties and predict the cellular locations of 69 *CaDSBT* and 82 *CaKSBT* proteins from desi and kabuli chickpea cultivars, respectively. Here, physicochemical properties including amino acid count, molecular weight, isoelectric point (pI), grand average of hydropathicity (GRAVY), and instability index of both desi and kabuli chickpea have been determined and are summarized in Table 2 and 3.

The amino acid numbers of SBT proteins were revealed a considerable variation, ranging from 116 to 1323 in desi and 165 to 1235 in kabuli chickpea with the average amino acid length approximately 636 amino acids in both selected cultivars. Similarly, the average molecular weight of SBT proteins was reported as 63 kDa for desi and 70 kDa for kabuli cultivars. The theoretical isoelectric points of SBT proteins were found to range from 4.7 to 10.1 across both chickpea cultivars. Interestingly, desi cultivar was reported with 25 acidic and 44 basic SBT proteins (out of 69 total), while kabuli cultivars with 36 acidic and 46 basic proteins (out

Table 1 List of primer pairs of SBT genes of desi and kabuli cultivars used for qRT-PCR.

Sr. No	Cultivar	Gene names	Primers pairs(5'-3')
1	Desi	CaDSBT59	CCAATCGCAGTCTCCAACAT CCTTTTGGCCCTTCCTCAA
2		CaDSBT11	TCTGGTGGTGCAATAGGACT TGGTTTGTCTCCCACTTCAG
3		CaDSBT8	AGTGCTGTGCAACCATTGAT GGAGCAGTTATGTCAGGCTG
4		CaDSBT57	TTGCGAGGGCTGTAAACAA CCATGTCCTCTACAAGCGTG
5		CaDSBT27	ACTCGAACACCAGAGTTCCT GATGTTGACGCCATCTTCGA
6		CaDSBT36	TGGCCTCTCGAAGATAGCAT GCTGAGTCGAAACCACAAGT
1	Kabuli	CaKSBT17	TGGAAAGGCAGTTGCATGAA AAGAACCGTTCTGTTGTGGG
2		CaKSBT71	TGCTGTGGATTAGAAGGGGT ACTCATGTGGTTCCCTCACT
3		CaKSBT14	CCTTTGTCTTCCCCCTCTCT GCGACAACACATGCAATAGC
4		CaKSBT11	CAGCTTCCAAGAGGCAGATC CGCTCCCATAGTCAAAAGGG
5		CaKSBT39	TGTGTAGGCATGAAGAGGGT CTGCTAGGATGTTCAACCCG
6		CaKSBT59	GTCCCATTCAGAGTCTCGC ATGGAGGGACTACTTTCGA

Table 2 The characteristics of SBT proteins in desi chickpea.

Protein name	Protein Id	Amino acids	Molecular weight (Da)	Isoelectric point (pI)	Gravy	Instability index	Subcellular localisation
CaDSBT_1	cicar.ICC4958.v2.0.Ca_01283.1	819	87,959.46	8.67	−0.047	27.73(stable)	Endoplasmic Reticulum
CaDSBT_2	cicar.ICC4958.v2.0.Ca_01342.1	852	91,413.76	7.32	0.004	34.28(stable)	Chloroplast
CaDSBT_3	cicar.ICC4958.v2.0.Ca_01446.1	774	82,935.31	6.83	−0.144	32.49(stable)	Extracellular
CaDSBT_4	cicar.ICC4958.v2.0.Ca_02475.1	773	84,745.14	8.61	−0.348	37.2(stable)	Cytoplasmic
CaDSBT_5	cicar.ICC4958.v2.0.Ca_02685.1	598	64,386.67	9.09	−0.033	31.63(stable)	Chloroplast
CaDSBT_6	cicar.ICC4958.v2.0.Ca_02921.1	784	undefined		−0.001	45.29(unstable)	Integral membrane protein
CaDSBT_7	cicar.ICC4958.v2.0.Ca_03157.1	667	72,532.02	6.51	−0.175	28.22(stable)	Cytoplasmic
CaDSBT_8	cicar.ICC4958.v2.0.Ca_03386.1	321	36,083.57	6.65	−0.297	41.17(unstable)	Nucleus
CaDSBT_9	cicar.ICC4958.v2.0.Ca_03930.1	318	34,138.01	5.76	0.173	37(stable)	Cytoplasmic
CaDSBT_10	cicar.ICC4958.v2.0.Ca_03963.1	287	31,157.59	8.33	−0.037	43.35(unstable)	Cytoplasmic
CaDSBT_11	cicar.ICC4958.v2.0.Ca_04364.1	771	83,789.61	8.96	−0.096	35.35(stable)	Extracellular
CaDSBT_12	cicar.ICC4958.v2.0.Ca_06496.1	457	48,906.7	7.15	−0.032	18.55(stable)	Chloroplast
CaDSBT_13	cicar.ICC4958.v2.0.Ca_06497.1	662	70,956.6	8.61	−0.126	30.61(stable)	Vacuolar
CaDSBT_14	cicar.ICC4958.v2.0.Ca_06546.1	672	73,049.72	6.25	−0.047	34.72(stable)	Cytoplasmic
CaDSBT_15	cicar.ICC4958.v2.0.Ca_06760.1	798	85,931.08	9.61	−0.118	29.88(stable)	Vacuolar
CaDSBT_16	cicar.ICC4958.v2.0.Ca_07089.1	777	84,996.78	6.32	−0.082	32.62(stable)	Chloroplast
CaDSBT_17	cicar.ICC4958.v2.0.Ca_07419.1	789	undefined		−0.171	35.8(stable)	Chloroplast
CaDSBT_18	cicar.ICC4958.v2.0.Ca_08229.1	243	27,777.32	4.76	−0.543	40.95(unstable)	Cytoplasmic
CaDSBT_19	cicar.ICC4958.v2.0.Ca_08891.1	683	73,436.79	8.71	−0.027	30.28(stable)	Cytoplasmic
CaDSBT_20	cicar.ICC4958.v2.0.Ca_09316.1	763	undefined		−0.335	37.99(stable)	Nucleus
CaDSBT_21	cicar.ICC4958.v2.0.Ca_10053.1	116	12,304.64	5.83	−0.467	26.44(stable)	Cytoplasmic
CaDSBT_22	cicar.ICC4958.v2.0.Ca_10312.1	1323	146,319.16	5.9	−0.337	38.59(stable)	Cytoplasmic
CaDSBT_23	cicar.ICC4958.v2.0.Ca_10346.1	786	85,097.24	8.83	−0.166	32.47(stable)	Cytoplasmic
CaDSBT_24	cicar.ICC4958.v2.0.Ca_13560.1	714	76,889.96	7.56	−0.253	34.73(stable)	Cytoplasmic
CaDSBT_25	cicar.ICC4958.v2.0.Ca_13646.1	144	16,110.64	9.61	−0.266	23.29(stable)	Cytoplasmic
CaDSBT_26	cicar.ICC4958.v2.0.Ca_14858.1	594	63,620.12	6.94	−0.07	31.3(stable)	Extracellular
CaDSBT_27	cicar.ICC4958.v2.0.Ca_14976.1	776	undefined		−0.419	52.21(stable)	Nucleus
CaDSBT_29	cicar.ICC4958.v2.0.Ca_16031.1	601	64,123.18	5.54	−0.1	30.54(stable)	Chloroplast
CaDSBT_30	cicar.ICC4958.v2.0.Ca_16284.1	856	91,796.79	6.08	−0.013	34.2(stable)	Endoplasmic Reticulum
CaDSBT_31	cicar.ICC4958.v2.0.Ca_17381.1	773	undefined		−0.042	26.55(stable)	Mitochondrial
CaDSBT_32	cicar.ICC4958.v2.0.Ca_17729.1	776	82,776.13	9.02	−0.042	38.16(stable)	Endoplasmic Reticulum
CaDSBT_33	cicar.ICC4958.v2.0.Ca_19218.1	663	undefined		0.06	39.9(stable)	Nucleus
CaDSBT_34	cicar.ICC4958.v2.0.Ca_19219.1	391	41,883.58	9.01	−0.071	27.92(stable)	Integral membrane protein
CaDSBT_35	cicar.ICC4958.v2.0.Ca_19716.1	723	77,764.54	6.41	−0.174	25.93(stable)	Chloroplast
CaDSBT_36	cicar.ICC4958.v2.0.Ca_20061.1	592	64,700.26	9.37	−0.186	33.26(stable)	Cytoplasmic
CaDSBT_37	cicar.ICC4958.v2.0.Ca_20097.1	249	26,547.58	9.18	0.019	21.02(stable)	Cytoplasmic
CaDSBT_38	cicar.ICC4958.v2.0.Ca_20816.1	676	72,932.66	6.52	−0.164	40.88(unstable)	Nucleus
CaDSBT_39	cicar.ICC4958.v2.0.Ca_21043.1	758	79,982.45	6.09	−0.012	28.61(stable)	Chloroplast
CaDSBT_40	cicar.ICC4958.v2.0.Ca_21063.1	753	80,115.93	7.31	−0.092	35.16(stable)	Cytoplasmic
CaDSBT_41	cicar.ICC4958.v2.0.Ca_21212.1	181	19,708.58	8.96	−0.447	47.79(unstable)	Chloroplast
CaDSBT_42	cicar.ICC4958.v2.0.Ca_21250.1	549	58,321.56	8.88	0.048	30.94(stable)	Nucleus
CaDSBT_43	cicar.ICC4958.v2.0.Ca_21254.1	774	82,974.85	6.14	−0.115	31.6(stable)	Vacuolar
CaDSBT_44	cicar.ICC4958.v2.0.Ca_21539.1	561	60,622.07	9.31	−0.119	25.3(stable)	Cytoplasmic
CaDSBT_45	cicar.ICC4958.v2.0.Ca_21851.1	817	87,501.93	7.13	−0.026	30.25(stable)	Endoplasmic Reticulum
CaDSBT_46	cicar.ICC4958.v2.0.Ca_21869.1	734	79,863.34	8.54	−0.144	36.35(stable)	Cytoplasmic
CaDSBT_47	cicar.ICC4958.v2.0.Ca_22006.1	753	80,436.61	6.17	−0.07	34.76(stable)	Chloroplast
CaDSBT_48	cicar.ICC4958.v2.0.Ca_22580.1	742	78,662.13	5.69	0.053	28.94(stable)	Chloroplast
CaDSBT_49	cicar.ICC4958.v2.0.Ca_22581.1	691	undefined		−0.015	31.86(stable)	Vacuolar
CaDSBT_50	cicar.ICC4958.v2.0.Ca_22582.1	763	81,922.94	5.24	−0.054	33.58(stable)	Endoplasmic Reticulum
CaDSBT_51	cicar.ICC4958.v2.0.Ca_22756.1	763	81,841.07	5.46	−0.014	34.32(stable)	Endoplasmic Reticulum
CaDSBT_52	cicar.ICC4958.v2.0.Ca_22756.1	723	77,191.74	8.43	−0.095	28.62(stable)	Chloroplast
CaDSBT_53	cicar.ICC4958.v2.0.Ca_22758.1	757	80,755.15	7.84	−0.001	28.58(stable)	Extracellular
CaDSBT_54	cicar.ICC4958.v2.0.Ca_23045.1	255	26,367.22	4.82	0.24	33.54(stable)	Cytoplasmic

Table 2 (continued)

Protein name	Protein Id	Amino acids	Molecular weight (Da)	Isoelectric point (pI)	Gravy	Instability index	Subcellular localisation
CaDSBT_55	cicar.ICC4958.v2.0.Ca_23047.1	127	14,322.74	4.87	0.21	32.97(stable)	Cytoplasmic
CaDSBT_56	cicar.ICC4958.v2.0.Ca_23049.1	1050	undefined		−0.237	39(stable)	Integral membrane protein
CaDSBT_57	cicar.ICC4958.v2.0.Ca_23258.1	157	17,269.58	7.59	−0.376	21.83(stable)	Nucleus
CaDSBT_58	cicar.ICC4958.v2.0.Ca_23523.1	763	82,067.05	8.58	−0.112	39.96(stable)	Chloroplast
CaDSBT_59	cicar.ICC4958.v2.0.Ca_24470.1	583	61,517.11	8.55	−0.047	28.36(stable)	Nucleus
CaDSBT_60	cicar.ICC4958.v2.0.Ca_24734.1	185	19,840.67	8.31	0.162	21.73(stable)	Chloroplast
CaDSBT_61	cicar.ICC4958.v2.0.Ca_25032.1	691	74,778.1	6.35	0.015	38.87(stable)	Chloroplast
CaDSBT_62	cicar.ICC4958.v2.0.Ca_25750.1	629	68,033.62	9.19	−0.19	34.28(stable)	Nucleus
CaDSBT_63	cicar.ICC4958.v2.0.Ca_26677.1	516	57,824.48	7.6	−0.105	43.62(unstable)	Nucleus
CaDSBT_64	cicar.ICC4958.v2.0.Ca_26786.1	805	undefined		−0.035	32.36(stable)	Extracellular
CaDSBT_65	cicar.ICC4958.v2.0.Ca_27437.1	466	51,053.32	8.69	−0.036	26.95(stable)	Cytoplasmic
CaDSBT_66	cicar.ICC4958.v2.0.Ca_27740.1	161	16,963.34	6.05	0.18	17.34(stable)	Cytoplasmic
CaDSBT_67	cicar.ICC4958.v2.0.Ca_27743.1	311	33,662.33	8.46	−0.207	39.02(stable)	Cytoplasmic
CaDSBT_68	cicar.ICC4958.v2.0.Ca_28133.1	380	41,412.74	10.1	−0.197	26.38	Nucleus
CaDSBT_69	cicar.ICC4958.v2.0.Ca_29811.1	762	82,078.51	9.13	−0.098	34.54	Chloroplast

of 82 total). In conclusion, majority of SBT proteins were found to be basic in nature. GRAVY analyses revealed that most SBT proteins in both cultivars are found to be hydrophilic in nature. GRAVY values were ranged from −0.543 to 0.24 in desi and −0.386 and 0.779 in kabuli indicating a prevalence of hydrophilic proteins. The instability index examination revealed that a significant number of curated SBT proteins in both desi (89.85%) and kabuli (89.02%) cultivars were found to be stable. Subcellular localization of SBT proteins revealed the distribution of SBT across diverse cellular compartments included the nucleus, vacuole, extracellular space, and endoplasmic reticulum. Approximately 25% of the SBT proteins were found to be localized in the chloroplast and 33% in the cytoplasm in both desi and kabuli chickpea cultivars.

Phylogenetic analysis reveals six distinct groups and differential distribution of *Subtilase* (SBT) genes in among chickpea cultivars

To examine the evolutionary dynamics of the *Subtilase* (SBT) gene family in chickpea, a comparative phylogenetic analysis was performed. A phylogenetic tree using SBT protein sequences, encompassing 69 CaDSBTs (desi) and 82 CaKSBTs (kabuli) sequences with 56 AtSBTs (*Arabidopsis thaliana*) and 122 AhSBTs (*Arachis hypogea*) protein sequences (Fig. 1 A and B). The phylogenetic analysis categorized the SBT proteins into six distinct groups (Groups 1–6). Group 1 was the largest clade in both chickpea cultivars, comprising 20 *CaDSBT* genes in desi and 24 *CaKSBT* genes in kabuli. Group 2 comprised 10 *CaDSBT* and 11 *CaKSBT* genes and Group 3 had fewer genes, with 3 *CaDSBT* and 2 *CaKSBT*. Group 4 contained with 20 *CaDSBT* and 21 *CaKSBT* genes,

while Group 5 included 6 *CaDSBT* and 18 *CaKSBT* genes. Interestingly, Group 6 contained significantly fewer genes in both cultivars, with only 3 *CaDSBTs* in desi and 2 *CaKSBTs* in kabuli.

Chromosome localization reveals SBT genes are dispersed across chickpea chromosomes, with a maximum on chromosome 7

A total of 69 *CaDSBT* and 82 *CaKSBT* genes were mapped across the eight chickpea chromosomes, exhibiting a largely random distribution pattern (Fig. 2 A and B). In desi chickpea genome, chromosome 7 (Chr. 7) reported with the highest number of genes, containing 14 genes. This was followed by chromosomes 2 and 3, each with 8 genes. Chromosomes 4 and 8 exhibited 7 genes each, while chromosome 5 reported with 6 genes. Chromosomes 1 and 6 each contained 5 genes. Similarly, in the kabuli chickpea, chromosome 7 also had the highest number of genes, containing 20 genes. It was followed by chromosome 3 with 12 genes and chromosome 2 with 10 genes. Chromosomes 6 and 8 each comprised 7 genes, whereas chromosome 4 had 6 genes, chromosome 5 had 5, and chromosome 1 had the least with 4 genes. Beyond the chromosomal mapping, 9 *CaDSBT* and 11 *CaKSBT* genes were found to be unevenly distributed across multiple scaffolds. *CaDSBT* genes were dispersed across distinct scaffolds, indicating that segmental duplication events likely contributed to their genomic expansion. Notably, certain *CaKSBT* genes formed clusters, such as *CaKSBT72-73* on Scaffold0867 and *CaKSBT75-76* on Scaffold1216.

Table 3 The characteristics of SBT proteins in Kabuli chickpea.

Protein name	Protein Id	Amino acids	Molecular weight (Da)	Isoelectric point (pI)	Gravy	Instability index	Subcellular localisation
CaKSBT_1	Ca_06276_v1.0_kabuli.1	761	81,788.94	9.16	-0.172	28.67(Stable)	Nucleus
CaKSBT_2	Ca_00451_v1.0_kabuli.1	773	84,745.14	8.61	-0.348	37.2(Stable)	Cytoplasmic
CaKSBT_3	Ca_00781_v1.0_kabuli.1	814	87,626.85	8.84	-0.154	31.24(Stable)	Nucleus
CaKSBT_4	Ca_01351_v1.0_kabuli.1	683	74,212.42	6.25	-0.109	32.17(Stable)	Cytoplasmic
CaKSBT_5	Ca_01584_v1.0_kabuli.1	747	80,393.55	8.69	-0.159	31.2(Stable)	Mitochondrial
CaKSBT_6	Ca_02197_v1.0_kabuli.1	757	80,755.15	7.84	-0.001	28.58(Stable)	Extracellular
CaKSBT_7	Ca_02199_v1.0_kabuli.1	723	77,191.74	8.43	-0.095	28.62(Stable)	Chloroplast
CaKSBT_8	Ca_02392_v1.0_kabuli.1	763	81,841.07	5.46	-0.014	34.32(Stable)	Endoplasmic Reticulum
CaKSBT_9	Ca_02393_v1.0_kabuli.1	763	81,922.94	5.24	-0.054	33.58(Stable)	Endoplasmic Reticulum
CaKSBT_10	Ca_03142_v1.0_kabuli.1	732	78,085.62	6.17	-0.132	35.42(Stable)	Mitochondrial
CaKSBT_11	Ca_03223_v1.0_kabuli.1	817	87,501.93	7.13	-0.026	30.25(Stable)	Endoplasmic Reticulum
CaKSBT_12	Ca_03395_v1.0_kabuli.1	742	78,662.13	5.69	0.053	28.94(Stable)	Cytoplasmic
CaKSBT_13	Ca_03817_v1.0_kabuli.1	683	73,436.79	8.71	-0.027	30.28(Stable)	Cytoplasmic
CaKSBT_14	Ca_04462_v1.0_kabuli.1	1235	136,505.28	5.61	-0.284	38.19(Stable)	Cytoplasmic
CaKSBT_15	Ca_04529_v1.0_kabuli.1	744	80,261.34	8.75	-0.217	32.11(Stable)	Cytoplasmic
CaKSBT_16	Ca_04530_v1.0_kabuli.1	714	76,889.96	7.56	-0.253	34.73(Stable)	Cytoplasmic
CaKSBT_17	Ca_04872_v1.0_kabuli.1	768	82,449.95	5.96	-0.118	32.46(Stable)	Cytoplasmic
CaKSBT_18	Ca_04873_v1.0_kabuli.1	776	82,677.09	6.17	-0.167	32.55(Stable)	Cytoplasmic
CaKSBT_19	Ca_05791_v1.0_kabuli.1	255	28,481.2	9.2	0.022	38.69(Stable)	Plastid
CaKSBT_20	Ca_06046_v1.0_kabuli.1	724	77,880.67	9.66	-0.178	31.23(Stable)	Chloroplast
CaKSBT_21	Ca_06049_v1.0_kabuli.1	724	77,795.61	9.69	-0.168	30.8(Stable)	Chloroplast
CaKSBT_22	Ca_06051_v1.0_kabuli.1	552	58,906.16	9.83	-0.076	28.74(Stable)	Chloroplast
CaKSBT_23	Ca_06053_v1.0_kabuli.1	787	84,949	9.63	-0.117	29.01(Stable)	Chloroplast
CaKSBT_24	Ca_06159_v1.0_kabuli.1	692	75,248.34	6.44	-0.071	35.59(Stable)	Cytoplasmic
CaKSBT_25	Ca_06277_v1.0_kabuli.1	717	76,923.89	9.02	-0.072	28.78(Stable)	Endoplasmic Reticulum
CaKSBT_26	Ca_07635_v1.0_kabuli.1	772	83,238.54	8.31	-0.129	30.71(Stable)	Chloroplast
CaKSBT_27	Ca_08442_v1.0_kabuli.1	178	18,650.93	4.57	0.022	25.03(Stable)	Nucleus
CaKSBT_28	Ca_08529_v1.0_kabuli.1	720	77,403.48	8.79	0.025	24.75(Stable)	Mitochondrial
CaKSBT_29	Ca_08725_v1.0_kabuli.1	753	80,115.93	7.31	-0.092	35.16(Stable)	Cytoplasmic
CaKSBT_30	Ca_08749_v1.0_kabuli.1	655	70,428.31	8.47	-0.018	31.05(Stable)	Cytoplasmic
CaKSBT_19	Ca_05791_v1.0_kabuli.1	255	28,481.2	9.2	0.022	38.69(Stable)	Plastid
CaKSBT_20	Ca_06046_v1.0_kabuli.1	724	77,880.67	9.66	-0.178	31.23(Stable)	Chloroplast
CaKSBT_21	Ca_06049_v1.0_kabuli.1	724	77,795.61	9.69	-0.168	30.8(Stable)	Chloroplast
CaKSBT_22	Ca_06051_v1.0_kabuli.1	552	58,906.16	9.83	-0.076	28.74(Stable)	Chloroplast
CaKSBT_23	Ca_06053_v1.0_kabuli.1	787	84,949	9.63	-0.117	29.01(Stable)	Chloroplast
CaKSBT_24	Ca_06159_v1.0_kabuli.1	692	75,248.34	6.44	-0.071	35.59(Stable)	Cytoplasmic
CaKSBT_25	Ca_06277_v1.0_kabuli.1	717	76,923.89	9.02	-0.072	28.78(Stable)	Endoplasmic Reticulum
CaKSBT_26	Ca_07635_v1.0_kabuli.1	772	83,238.54	8.31	-0.129	30.71(Stable)	Chloroplast
CaKSBT_27	Ca_08442_v1.0_kabuli.1	178	18,650.93	4.57	0.022	25.03(Stable)	Nucleus
CaKSBT_28	Ca_08529_v1.0_kabuli.1	720	77,403.48	8.79	0.025	24.75(Stable)	Mitochondrial
CaKSBT_29	Ca_08725_v1.0_kabuli.1	753	80,115.93	7.31	-0.092	35.16(Stable)	Cytoplasmic
CaKSBT_30	Ca_08749_v1.0_kabuli.1	655	70,428.31	8.47	-0.018	31.05(Stable)	Cytoplasmic
CaKSBT_31	Ca_08752_v1.0_kabuli.1	743	81,040.26	8.99	-0.161	28.51(Stable)	Chloroplast
CaKSBT_32	Ca_08754_v1.0_kabuli.1	599	63,878.42	8.25	-0.048	23.98(Stable)	Chloroplast
CaKSBT_33	Ca_08755_v1.0_kabuli.1	778	83,460.45	6.14	-0.123	31.3(Stable)	Vacuolar
CaKSBT_34	Ca_08848_v1.0_kabuli.1	819	87,959.46	8.67	-0.047	27.73(Stable)	Endoplasmic Reticulum
CaKSBT_35	Ca_09331_v1.0_kabuli.1	758	79,982.45	6.09	-0.012	28.61(Stable)	Chloroplast
CaKSBT_36	Ca_09754_v1.0_kabuli.1	779	85,024.27	6.81	-0.148	31.66(Stable)	Extracellular
CaKSBT_37	Ca_09859_v1.0_kabuli.1	762	82,078.51	9.13	-0.098	34.54(Stable)	Chloroplast
CaKSBT_38	Ca_10295_v1.0_kabuli.1	539	57,680.61	6.33	0.013	27.49(Stable)	Chloroplast
CaKSBT_39	Ca_10694_v1.0_kabuli.1	763	82,067.05	8.58	-0.112	39.96(Stable)	Chloroplast
CaKSBT_40	Ca_11029_v1.0_kabuli.1	776	82,776.13	9.02	-0.042	38.16(Stable)	Endoplasmic Reticulum
CaKSBT_41	Ca_12270_v1.0_kabuli.1	720	78,099.26	9.26	-0.14	34.67(Stable)	Chloroplast

Table 3 (continued)

Protein name	Protein Id	Amino acids	Molecular weight (Da)	Isoelectric point (pI)	Gravy	Instability index	Subcellular localisation
CaKSBT_42	Ca_12959_v1.0_kabuli.1	985	110,319.85	5.73	-0.269	42.42(Unstable)	Plastid
CaKSBT_43	Ca_14938_v1.0_kabuli.1	230	23,889.36	4.77	0.245	32.72(Unstable)	Cytoplasmic
CaKSBT_44	Ca_16079_v1.0_kabuli.1	662	71,243.88	7.92	-0.157	40.13(Unstable)	Mitochondrial
CaKSBT_45	Ca_16081_v1.0_kabuli.1	674	72,534.54	6.65	-0.055	35.33(Stable)	Extracellular
CaKSBT_46	Ca_16143_v1.0_kabuli.1	788			0.779	28.72(Stable)	Vacuolar
CaKSBT_47	Ca_16181_v1.0_kabuli.1	753	79,823.6	8.75	-0.066	33.59(Stable)	Cytoplasmic
CaKSBT_48	Ca_16421_v1.0_kabuli.1	650	69,942.95	9.17	-0.057	31.95(Stable)	Chloroplast
CaKSBT_49	Ca_16470_v1.0_kabuli.1	698			-0.42	43(Unstable)	Nucleus
CaKSBT_50	Ca_16543_v1.0_kabuli.1	207	21,516.25	4.82	0.065	29.07(Stable)	Nucleus
CaKSBT_51	Ca_16746_v1.0_kabuli.1	845	90,634.37	6.02	-0.021	34.7(Stable)	Endoplasmic Reticulum
CaKSBT_52	Ca_17668_v1.0_kabuli.1	743	80,656.81	8.89	-0.155	36.15(Stable)	Peroxisomal
CaKSBT_53	Ca_17808_v1.0_kabuli.1	763	82,388.83	5.83	-0.145	29.09(Stable)	Cytoplasmic
CaKSBT_54	Ca_18162_v1.0_kabuli.1	757	82,372.64	7.64	-0.168	36.38(Stable)	Chloroplast
CaKSBT_55	Ca_18358_v1.0_kabuli.1	717	76,907.19	6.94	-0.113	34.1(Stable)	Chloroplast
CaKSBT_56	Ca_18597_v1.0_kabuli.1	837	89,701.6	7.05	-0.016	34.54(Stable)	Cytoplasmic
CaKSBT_57	Ca_19955_v1.0_kabuli.1	676	72,932.66	6.52	-0.164	40.88(Unstable)	Nucleus
CaKSBT_58	Ca_21657_v1.0_kabuli.1	327	35,872.97	6.86	-0.07	36.91(Stable)	Cytoplasmic
CaKSBT_59	Ca_21726_v1.0_kabuli.1	599	64,677.69	9.1	-0.169	32.22(Stable)	Cytoplasmic
CaKSBT_60	Ca_21732_v1.0_kabuli.1	692	75,434.05	8.92	-0.204	36.6(Stable)	Cytoplasmic
CaKSBT_61	Ca_22360_v1.0_kabuli.1	744	79,604.34	6.53	-0.188	32.31(Stable)	Cytoplasmic
CaKSBT_62	Ca_22421_v1.0_kabuli.1	676	73,333.59	9.07	-0.124	26.23(Stable)	Chloroplast
CaKSBT_63	Ca_22422_v1.0_kabuli.1	651	69,786.63	6.15	-0.146	24.91(Stable)	Cytoplasmic
CaKSBT_64	Ca_23388_v1.0_kabuli.1	521	55,292.62	6.86	-0.137	29.69(Stable)	Chloroplast
CaKSBT_65	Ca_24492_v1.0_kabuli.1	176	19,507.95	4.7	-0.386	21.03(Stable)	Cytoplasmic
CaKSBT_66	Ca_24900_v1.0_kabuli.1	874	95,687.08	6.53	-0.329	39.59(Stable)	Nucleus
CaKSBT_67	Ca_25318_v1.0_kabuli.1	600	64,572.59	7.94	-0.018	32.92(Stable)	Cytoplasmic
CaKSBT_68	Ca_25596_v1.0_kabuli.1	165	17,972.46	8.82	-0.164	28.62(Stable)	Chloroplast
CaKSBT_69	Ca_25737_v1.0_kabuli.1	652	70,066.5	6.53	-0.065	36.76(Stable)	Cytoplasmic
CaKSBT_70	Ca_25817_v1.0_kabuli.1	619	67,329.92	6.44	0.022	28.64(Stable)	Cytoplasmic
CaKSBT_71	Ca_25888_v1.0_kabuli.1	924	102,144.38	7.53	-0.386	48.77(Unstable)	Chloroplast
CaKSBT_72	Ca_26384_v1.0_kabuli.1	674	73,355.42	6.89	-0.071	28.76(Stable)	Cytoplasmic
CaKSBT_73	Ca_26388_v1.0_kabuli.1	292	32,011.82	9.66	-0.236	37.85(Stable)	Chloroplast
CaKSBT_74	Ca_26475_v1.0_kabuli.1	287	31,157.59	8.33	-0.037	43.35(Unstable)	Cytoplasmic
CaKSBT_75	Ca_26928_v1.0_kabuli.1	228	24,859.63	8.54	-0.279	40.71(Unstable)	Chloroplast
CaKSBT_76	Ca_26929_v1.0_kabuli.1	401	45,470.1	5.42	-0.173	45.36(Unstable)	Cytoplasmic
CaKSBT_77	Ca_27263_v1.0_kabuli.1	290	31,713.13	8.75	-0.183	30.34(Stable)	Chloroplast
CaKSBT_78	Ca_27434_v1.0_kabuli.1	754	82,199.33	6.51	-0.096	34.29(Stable)	Integral membrane protein
CaKSBT_73	Ca_26388_v1.0_kabuli.1	292	32,011.82	9.66	-0.236	37.85(Stable)	Chloroplast
CaKSBT_74	Ca_26475_v1.0_kabuli.1	287	31,157.59	8.33	-0.037	43.35(Unstable)	Cytoplasmic
CaKSBT_75	Ca_26928_v1.0_kabuli.1	228	24,859.63	8.54	-0.279	40.71(Unstable)	Chloroplast
CaKSBT_76	Ca_26929_v1.0_kabuli.1	401	45,470.1	5.42	-0.173	45.36(Unstable)	Cytoplasmic
CaKSBT_77	Ca_27263_v1.0_kabuli.1	290	31,713.13	8.75	-0.183	30.34(Stable)	Chloroplast
CaKSBT_78	Ca_27434_v1.0_kabuli.1	754	82,199.33	6.51	-0.096	34.29(Stable)	Integral membrane protein
CaKSBT_79	Ca_27477_v1.0_kabuli.1	236	25,030.57	6.06	0.16	21.73(Stable)	Cytoplasmic
CaKSBT_80	Ca_27788_v1.0_kabuli.1	234	25,393.25	8.97	0.139	31.16(Stable)	Chloroplast
CaKSBT_81	Ca_28025_v1.0_kabuli.1	671	72,625.96	6.71	-0.009	31.56(Stable)	Cytoplasmic
CaKSBT_82	Ca_28164_v1.0_kabuli.1	721	78,319.29	6.61	-0.063	34.89(Stable)	Chloroplast

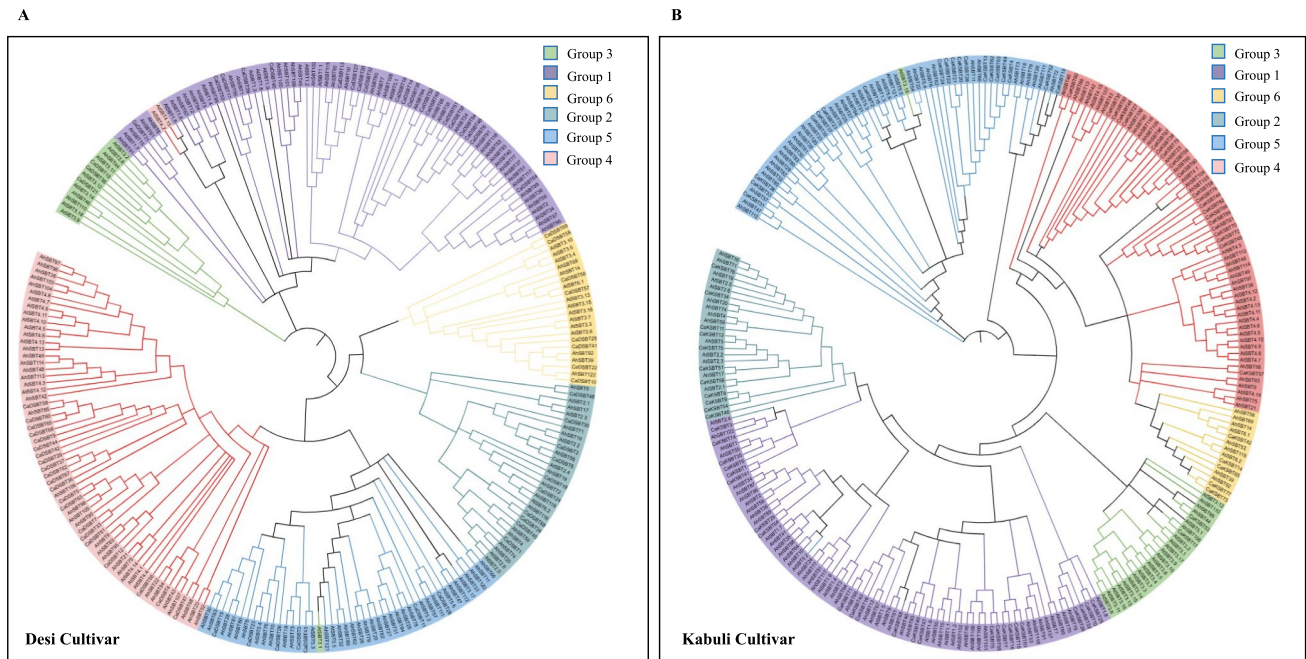


Fig. 1 Phylogenetic analysis of *C. arietinum* SBT genes. **a** Phylogenetic tree illustrating the relationships among the *CaDSBTs* gene family and previously identified *SBT* genes from *Arabidopsis thaliana* (*AtSBTs*) and *Arachis hypogaea* (*AhSBTs*). **b** Phylogenetic tree showing the relationships among the *CaKSBTs* gene family and previously

identified *SBT* genes from *Arabidopsis thaliana* (*AtSBTs*) and *Arachis hypogaea* (*AhSBTs*). Both phylogenetic trees were constructed using MEGA 11, with the six major groups highlighted by distinct color backgrounds

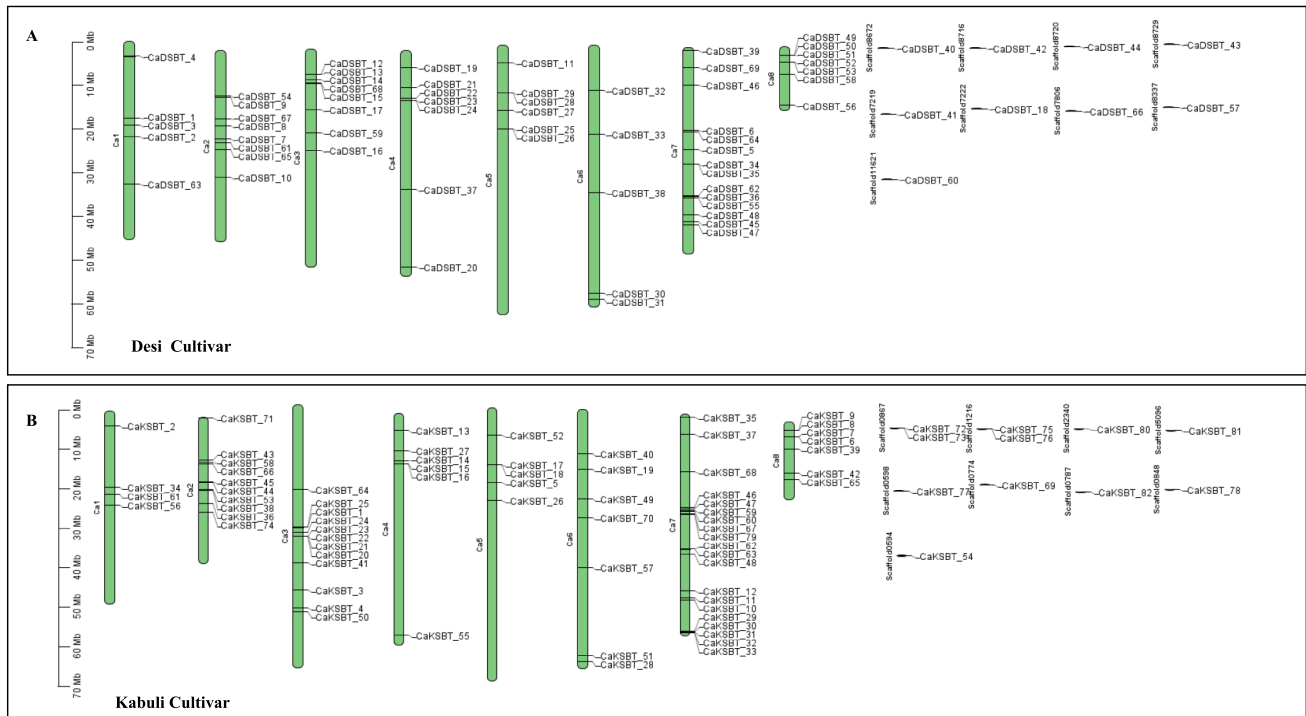


Fig. 2 Chromosomal distribution of *C. arietinum* SBT genes. **a** Distribution of *CaDSBT* genes (desi type) across chromosomes. **b** Distribution of *CaKSBT* genes (kabuli type) across chromosomes

Gene structure analysis revealed Group 1 contained the majority of intronless genes

The *CaDSBT* and *CaKSBT* genes in chickpea were revealed substantial variation in gene length, ranging from 3000 to 18,000 base pairs (bp). An in-depth analysis of their exon–intron structures was conducted using the Gene Structure Display Server (GSDS) to elucidate their genomic organization. In the desi chickpea genome, the number of exons per gene varied from 1 to 33, while introns ranged from 0 to 32 (Fig. 3 A). Notably, 19 *CaDSBT* genes were found to be intronless and 7 *CaDSBT* genes found to contain a single intron. The *CaDSBT22* gene was exhibited maximum of 32 introns. Most of the genes with no intron were from group 1, 12 genes (*CaDSBT50*, *CaDSBT51*, *CaDSBT6*, *CaDSBT27*, *CaDSBT64*, *CaDSBT20*, *CaDSBT32*, *CaDSBT53*, *CaDSBT17*, *CaDSBT39*, *CaDSBT49*, *CaDSBT52*). Similarly, in the kabuli chickpea genome, number of exons ranged from 1 to 31, and intron numbers varied from 0 to 30. A total of 15 *CaKSBT* genes were identified as intronless, while 10 other *CaKSBT* genes contained a single intron (Fig. 3 B). The *CaKSBT14* gene was found to the maximum intron number at 30. Consistent with the desi cultivar, most intronless genes in kabuli chickpea also clustered within Group 1, including 8 genes (*CaKSBT6*, *CaKSBT35*, *CaKSBT1*, *CaKSBT29*, *CaKSBT40*, *CaKSBT15*, *CaKSBT16*, and *CaKSBT7*). Analysis of the domain architecture for both

CaDSBT and *CaKSBT* genes revealed a largely conserved four-domain structure. For *CaDSBT* proteins, the central region is characterized by the presence of a peptidase S8_S53 superfamily domain. At the N-terminal, an inhibitor_I9 prodomain was identified in 54% of the *CaDSBT* proteins, while a fn3_6 domain was consistently found at the C-terminal in 65% of the proteins. Interestingly, 33 *CaDSBT* genes (48%) were found to lack the insertion consistent with the PA domain. Similarly, *CaKSBT* also exhibited four-domain structure comparable to their *CaDSBT*. These include the central peptidase S8_S53 superfamily domain and the associated PA domain. An inhibitor_I9 prodomain was present at the N-terminal in 56% of *CaKSBT* proteins, and a fn3_6 domain was observed at the C-terminal in 78% of the proteins. A higher proportion of *CaKSBT* genes, specifically 53 genes (65%), were found to lack the insertion consistent with the PA domain. In conclusion, both conserved structural elements and variations in specific domain occurrences, particularly the PA domain, between the *CaDSBT* and *CaKSBT* gene families in chickpea.

Motif analysis of chickpea SBT proteins reveals cultivar-specific motif distributions

Motif composition of the curated chickpea subtilase (SBT) proteins, including 69 *CaDSBTs* and 82 *CaKSBTs*, was performed using MEME software to understand the functional

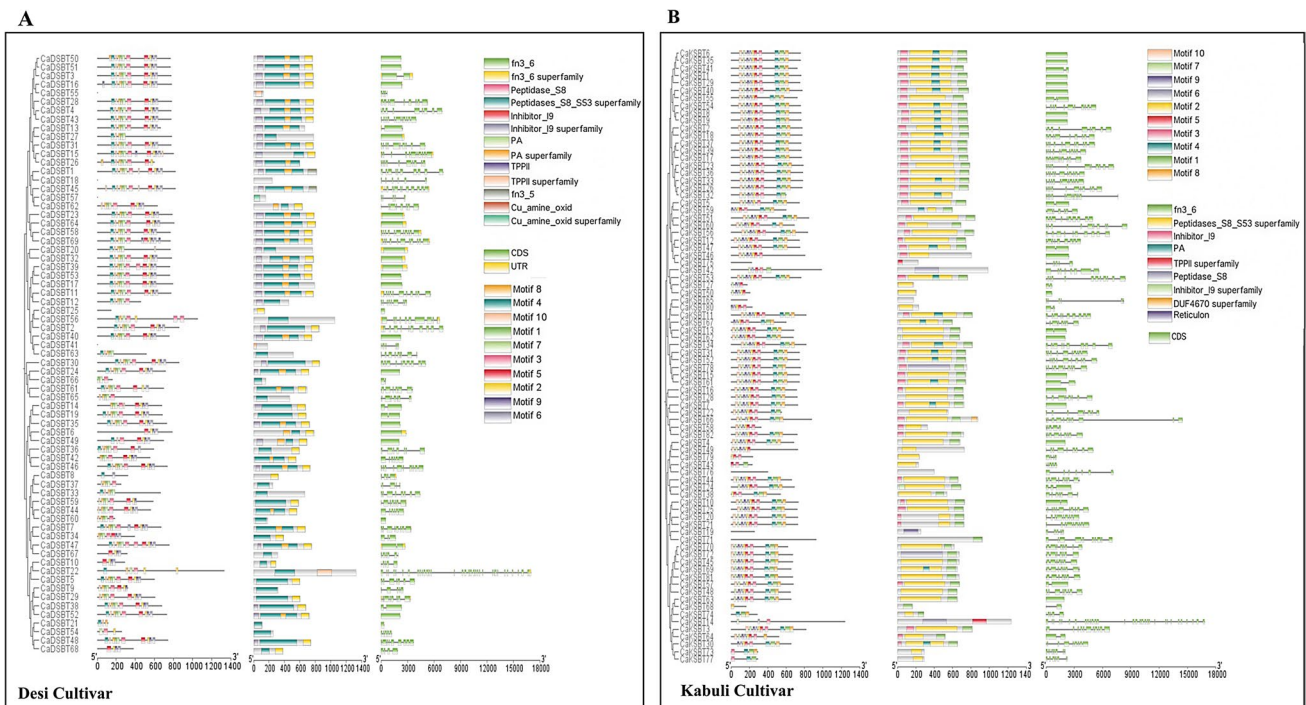


Fig. 3 Gene structure and intron–exon organization of *SBT* genes. **a** Intron–exon structure and motif arrangement of *CaDSBT* genes. **b** Intron–exon structure and motif arrangement of *CaKSBT* genes. Neighbor-joining trees (1000 bootstraps) identified 10 conserved

motifs (color-coded). Coding regions (CDS) are shown as yellow boxes, untranslated regions (UTR) as green boxes, and introns as black lines. The scale indicates feature location from 5' to 3'

characterisation of SBT gene family (Fig. 4 A and B). A total of 64 CaDSBTs and 77 CaKSBTs were found to possess 10 distinct types of motifs. Interestingly, five remaining proteins in both desi (CaDSBT55, CaDSBT18, CaDSBT57, CaDSBT25, CaDSBT41) and kabuli (CaKSBT75, CaKSBT65, CaKSBT76, CaKSBT19, CaKSBT71) cultivars lacked any identifiable motifs. The absence of identifiable motifs in some chickpea (SBT) protein is likely due to the combination of factors related to their structure and evolutionary divergence. These factors could be the protein size and sequence diversity, motif localization and intrinsically disordered proteins. In conclusion, cultivar-specific motif distributions within the chickpea SBT protein family, suggesting potential functional divergences and cultivar specific.

Ka/Ks ratio analysis found purifying selection as dominant evolutionary force for the SBT gene family chickpea

The Ka/Ks ratio, comprises the rate of non-synonymous (Ka) to synonymous (Ks) substitutions, was employed to determine the selective pressures of *CaDSBT* and *CaKSBT* gene pairs in chickpea. The Ka, Ks, and the Ka/Ks ratio values have been provided in Table 4. In desi cultivars, 14 *CaDSBT* gene pairs were identified with Ka/Ks values were consistently less than 1, ranging from 0.143 to 0.739 indicating genes have primarily evolved under purifying (negative) selection, a process that removes deleterious mutations

and preserves gene function and suggested adaptive evolution. The estimated divergence times for these *CaDSBT* gene duplications range from approximately 5.4 to 144 million years ago (MYA). In kabuli chickpea, 24 *CaKSBT* gene pairs were identified with majority having Ka/Ks ratios less than 1, signifying purifying (negative) selection. In contrast, one gene pair (*CaKSBT27* and *CaKSBT50*) was exhibited a Ka/Ks ratio greater than 1, indicative of positive selection. The Ka/Ks ratios for *CaKSBTs* ranged from 0.121 to 1.226, suggesting a broader spectrum of duplication events, from very recent (approximately 0.00066 MYA or ~658 years ago) to ancient (up to 361.22 MYA). In-fact, the gene pairs with all reported Ka/Ks ratio of 1 does not rule out neofunctionalization particularly through expression divergence. In conclusion, purifying selection has been the predominant evolutionary force for the *Subtilase* (SBT) gene family in both desi and kabuli cultivars of *C. arietinum*.

Gene ontology revealed SBT proteins primarily function as serine-type endopeptidases in chickpea

Gene Ontology (GO) analysis was performed to elucidate the diverse functional roles of the chickpea Subtilase (SBT) proteins, categorizing their involvement in biological processes, molecular functions, and cellular components (Fig. 5 A–F). In terms of biological processes, SBT proteins in desi chickpea showed maximum involvement in proteolysis, indicating their significant role in protein degradation pathways. Conversely, SBT proteins in kabuli chickpea

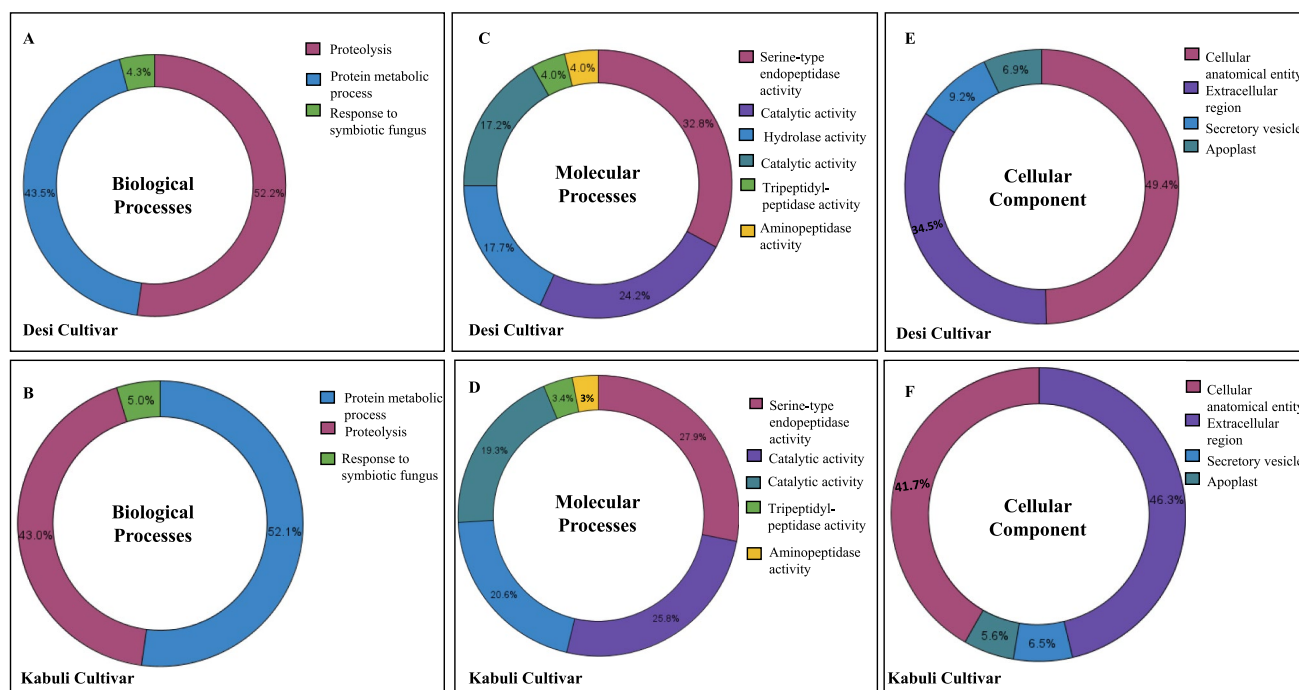


Fig. 4 Conserved motifs in SBT proteins identified by MEME. **a** Conserved Motifs of CaDSBT proteins (desi). **b** Conserved Motifs of CaKSBT proteins (kabuli). Letter size reflects amino acid conservation within each motif; larger letters indicate higher conservation

Table 4 Ka, Ks, Ka/Ks ratio, and divergent time of the duplicated SBT genes in Chickpea plants.

Cultivar	Genes1	Genes2	Ka	Ks	Ka/Ks	MYA ¹
Desi	CaDSBT_50	CaDSBT_51	0.04847067498	0.06551016838	0.7398954418	5.36
	CaDSBT_4	CaDSBT_43	0.5672849069	1.475342061	0.3845107667	120.92
	CaDSBT_58	CaDSBT_69	0.1277344818	0.5824840554	0.2192926666	47.74
	CaDSBT_39	CaDSBT_53	0.2513211195	1.754577804	0.1432373753	143.81
	CaDSBT_41	CaDSBT_63	1.313255832	2.801720037	0.4687319983	229.64
	CaDSBT_14	CaDSBT_19	0.2222374188	0.6398259411	0.3473404321	52.44
	CaDSBT_42	CaDSBT_46	0.3385108204	1.522748046	0.2223025807	124.81
	CaDSBT_44	CaDSBT_60	0.6312240921	1.553934956	0.4062101116	127.37
	CaDSBT_7	CaDSBT_34	0.5252486915	1.39179834	0.3773885027	114.08
	CaDSBT_47	CaDSBT_67	0.6854526977	3.091560988	0.2217173461	253.4
	CaDSBT_10	CaDSBT_22	0.883809993	2.08758173	0.4233654569	171.11
	CaDSBT_9	CaDSBT_29	0.5608624973	2.891022383	0.1940014372	236.96
	CaDSBT_38	CaDSBT_52	0.565362644	2.477584199	0.2281910921	203.08
	CaDSBT_48	CaDSBT_68	0.7446611639	2.005360357	0.371335337	164.37
Kabuli	CaKSBT_6	CaKSBT_35	0.2513211195	1.754577804	0.1432373753	143.81
	CaKSBT_40	CaKSBT_55	0.4692955383	2.376915724	0.1974388631	194.82
	CaKSBT_8	CaKSBT_9	0.04847067498	0.06551016838	0.7398954418	5.36
	CaKSBT_2	CaKSBT_18	0.5773426536	2.38812902	0.2417552187	195.74
	CaKSBT_37	CaKSBT_39	0.1277344818	0.5824840554	0.2192926666	47.74
	CaKSBT_15	CaKSBT_61	0.4705741747	3.014845804	0.1560856525	247.11
	CaKSBT_58	CaKSBT_82	0.2856393229	1.003755941	0.2845704929	82.27
	CaKSBT_24	CaKSBT_38	0.6036845116	1.65159131	0.3655168854	135.37
	CaKSBT_19	CaKSBT_71	0.8564817943	2.423159227	0.3534566712	198.61
	CaKSBT_3	CaKSBT_64	0.5787668051	2.582765287	0.2240880377	211.7
	CaKSBT_23	CaKSBT_36	0.3485501342	0.9288662555	0.3752425412	76.13
	CaKSBT_26	CaKSBT_32	0.3623613036	1.270408028	0.2852322211	104.13
	CaKSBT_5	CaKSBT_59	0.5906421242	1.97025862	0.2997789824	161.49
	CaKSBT_51	CaKSBT_60	0.7599018135	2.505350652	0.3033115596	205.35
	CaKSBT_16	CaKSBT_28	0.5220508325	1.671946698	0.312241313	137.04
	CaKSBT_70	CaKSBT_72	0.03331065148	0.06814861117	0.4887942822	5.58
	CaKSBT_73	CaKSBT_77	0.2184934604	0.7857595647	0.2780665616	64.4
	CaKSBT_27	CaKSBT_50	0.00984831549	0.008032205285	1.22610356	0.65
	CaKSBT_13	CaKSBT_62	0.4257558534	1.210623493	0.3516831253	99.23
	CaKSBT_31	CaKSBT_52	0.3360910824	1.177718625	0.2853746857	96.53
	CaKSBT_7	CaKSBT_22	0.5343378306	4.406973815	0.1212482427	361.22
	CaKSBT_43	CaKSBT_76	0.9637150307	2.944210824	0.3273254153	241.32
	CaKSBT_20	CaKSBT_21	0.009854381143	0.02074299404	0.4750703357	1.7
	CaKSBT_63	CaKSBT_68	0.6233219144	2.503604434	0.2489698076	205.21

Million year ago¹.

were predominantly associated with protein metabolic processes, suggesting a broader involvement in various aspects of protein anabolism and catabolism. In context of molecular functions, serine-type endopeptidase activity was found as the dominant function for SBT proteins in both desi and kabuli chickpea cultivars. For cellular localization, cellular anatomical entities were identified as the primary cellular components for SBTs in desi chickpea. In contrast, extracellular regions were the major documented cellular components for SBTs in kabuli chickpea.

Cis-regulatory element analysis revealed SBT with significant potential for abiotic stress and phytohormone responses

Cis-regulatory element (CRE) analysis of *CaDSBT* and *CaKSBT* promoters revealed distinct patterns in their responsiveness to phytohormones, abiotic stress, and developmental cues (Fig. 6 A and B). Abiotic stresses CREs were found more in kabuli (1141) than desi (924), with light-responsive elements being most abundant in both cultivars (Supplementary Table 1). Kabuli also showed more defense and stress-related elements, while desi had slightly more drought-responsive elements. For development and growth

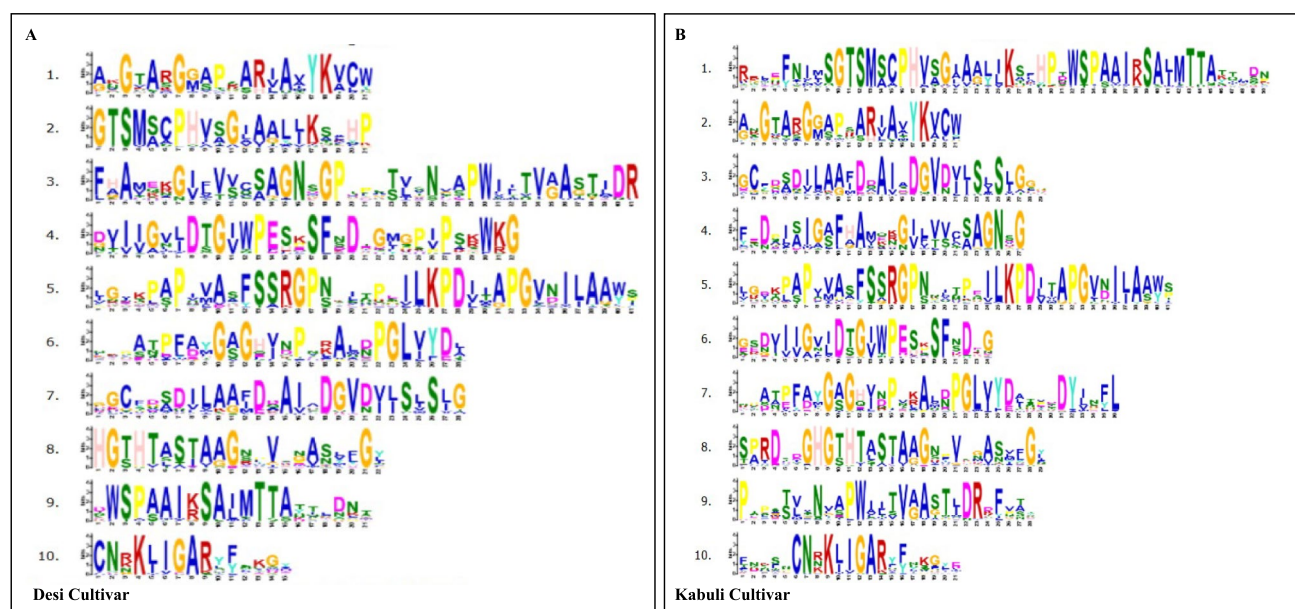


Fig. 5 Gene Ontology (GO) enrichment analysis of *SBT* genes using Blast2GO. **a** Biological processes (desi). **b** Biological processes (kabuli). **c** Cellular processes (desi). **d** Molecular processes (kabuli). **e** Cellular components (desi). **f** Cellular components (kabuli)

CREs Zein metabolism elements were most prevalent. Kabuli displayed higher numbers of meristem and endosperm expression elements, suggesting stronger regulatory influence in these processes, while desi had more circadian control elements. Phytohormone-responsive CREs were more frequent in kabuli than desi. Although gibberellin elements were equal in both cultivars, Kabuli exhibited higher frequencies of elements associated with salicylic acid, MeJA, abscisic acid, and auxin, indicating a potentially enhanced and dynamic hormone-responsive regulatory potential in kabuli *SBT* genes, particularly for stress-related signaling.

Protein structure analysis of *SBT* genes show cultivar-specific variations in secondary structure

The protein structures of six selected Subtilase (*SBT*) genes in desi and kabuli chickpea was reported with considerable variation in their secondary structure (Fig. 7 A and B). In desi cultivars, alpha-helix was ranged from 9 to 19%, while the beta-strand content was higher, ranging from 24 to 38% suggesting a predominant β -sheet architecture within these proteins and subsequently enhance protein's stability against stress induced proteolysis. Disordered regions were found to be varied from 2 to 18%. In context of majority of *SBT* proteins were predicted to lack transmembrane helices, with only CaDSBT27 (2%) and CaDSBT59 (3%) containing minimal percentages. In kabuli cultivars, the alpha-helix content was found to range from 9 to 15%, while the beta-strand content ranged from 24 to 31%, indicating a more balanced representation of both secondary structures. Disordered regions were generally low, ranging from 0 to 8% for

most genes in kabuli cultivars. Minimal levels of transmembrane helices (2%) were detected in four kabuli *SBT* genes, hinting at potential, albeit limited, membrane association for these proteins.

Chickpea cultivars show morphological changes, reduced RWC, increased ELI, and elevated MDA content under drought and cold stress

Seven-day-old chickpea plantlets exhibited distinct morphological responses to increasing concentrations of polyethylene glycol (PEG-6000)-induced drought stress and to cold stress (Fig. 8 A and B). Drought stress induced a progressive intensification of symptoms correlating with increased PEG-6000 concentrations. At a concentration of 10% (w/v) PEG-6000, plantlets displayed initial signs of stress, characterized by mild wilting and a slight reduction in leaf expansion. Increasing the PEG-6000 concentration to 20% (w/v) resulted in more pronounced symptoms, including intensified leaf rolling and a noticeable dull green discoloration of the foliage. Under severe drought conditions, specifically at 30% (w/v) PEG-6000, plantlets exhibited severe dehydration symptoms and a significant reduction in overall plant height when compared to control plantlets. In contrast to drought stress, plantlets subjected to cold stress (4 °C) displayed different, albeit less severe, morphological changes. These included slight leaf curling and a reduced overall vigor when compared to the control group. The control group remained healthy throughout the observation period, characterized by fully expanded, vibrant green leaves and upright stems, indicating optimal growth conditions.

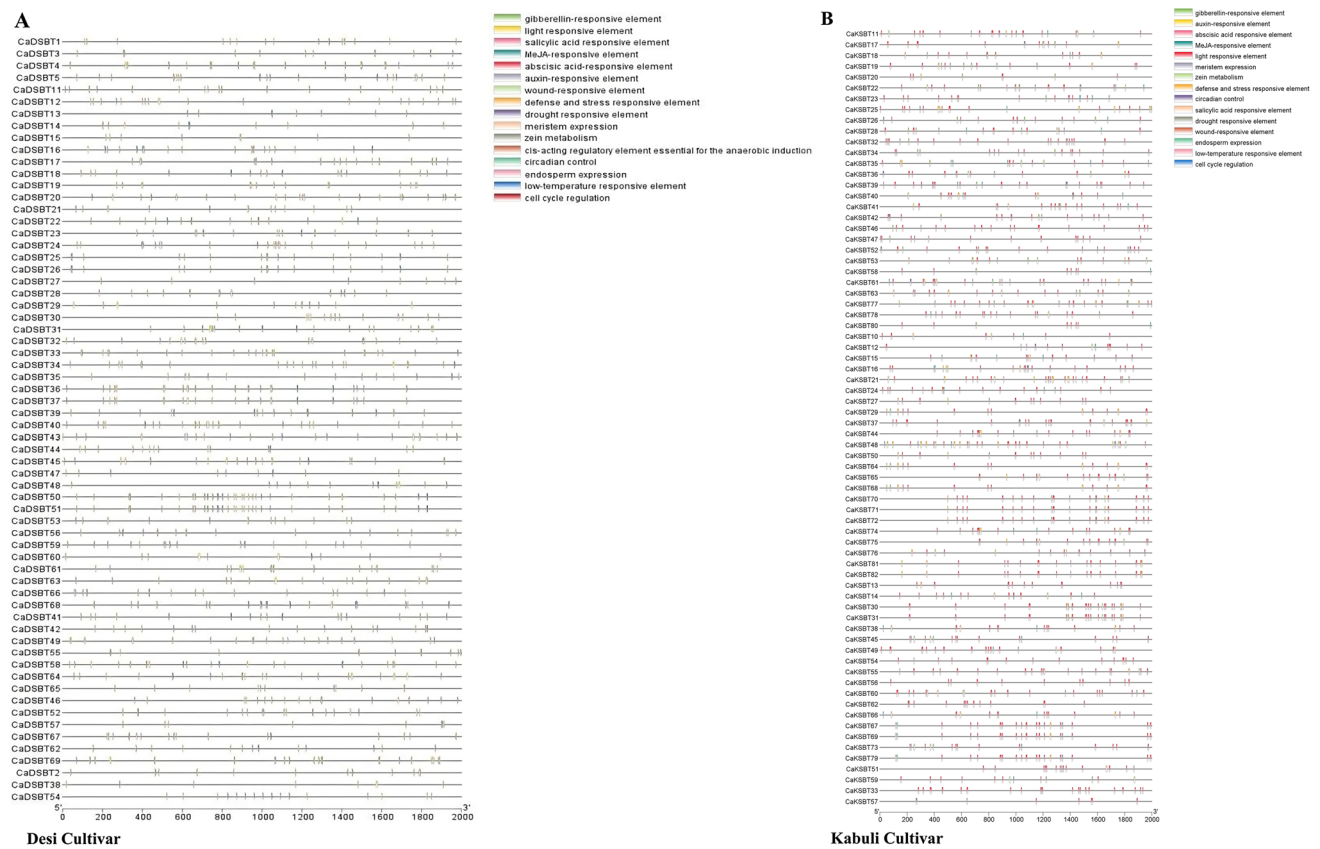


Fig. 6 Promoter analysis of SBT genes **a** *CaDSBT* (desi). **b** *CaKSBT* (kabuli)

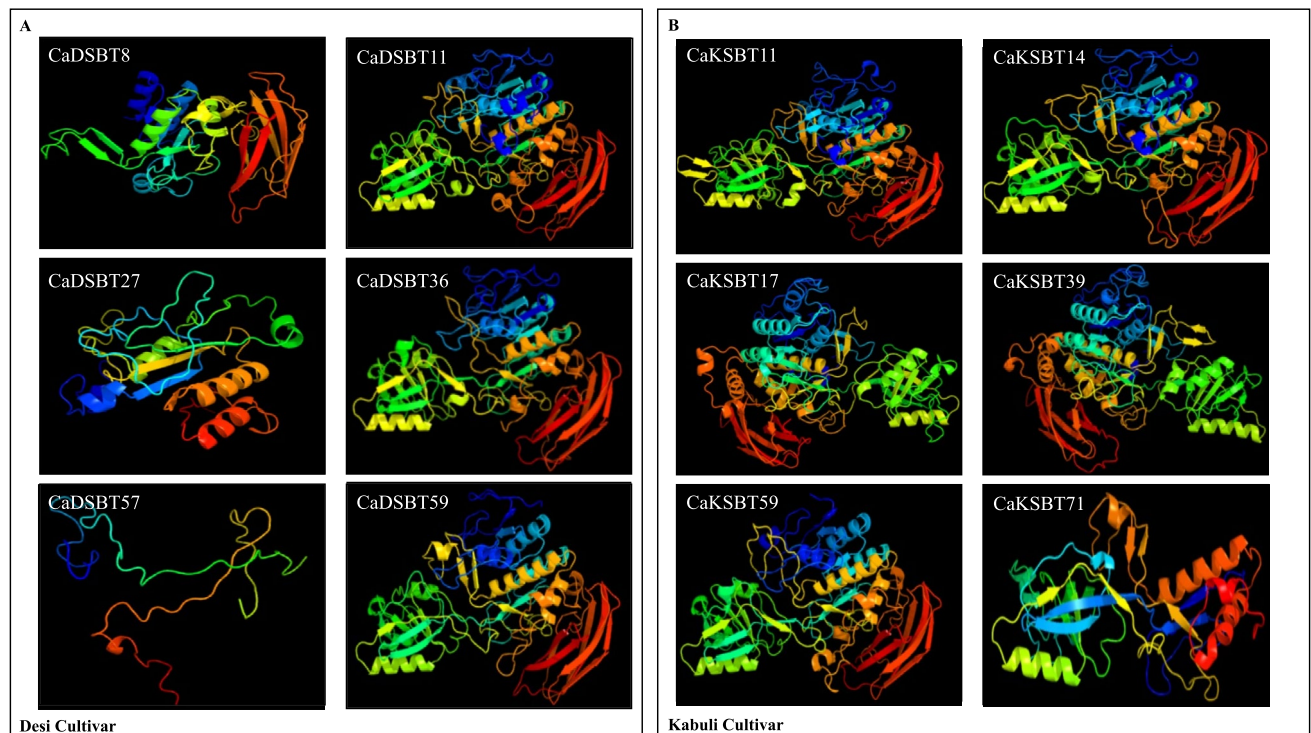


Fig. 7 Predicted three-dimensional structures of *C. arietinum* SBT proteins. **a** Structures of CaDSBT proteins. **b** Structures of CaKSBT proteins

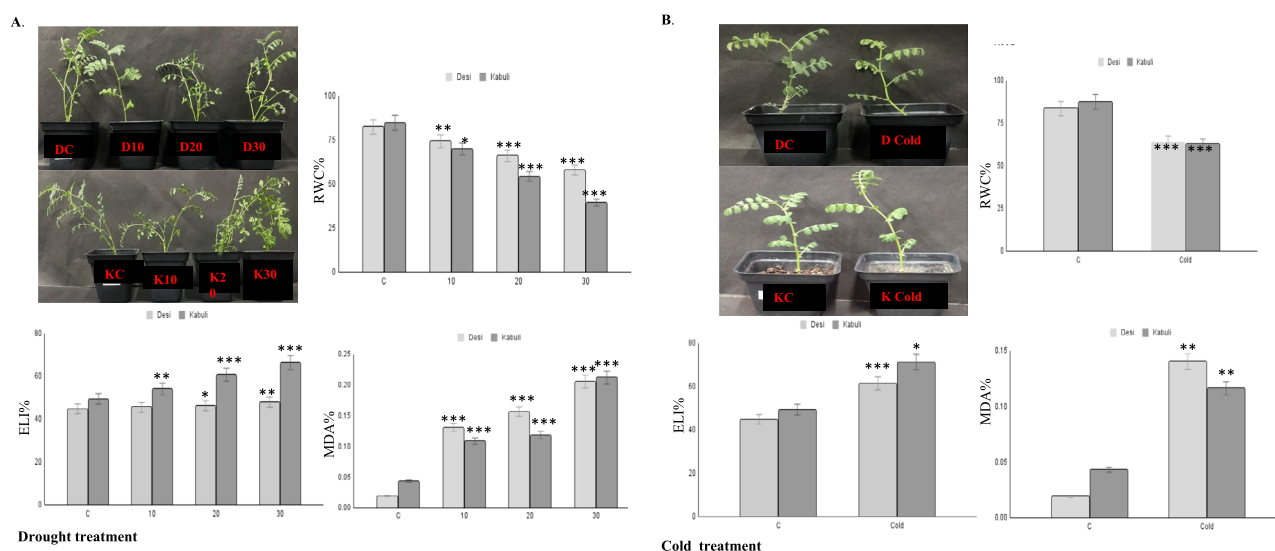


Fig. 8 Physiological analysis of desi and kabuli chickpea cultivars under drought and cold stress conditions. **a** Chickpea plants exposed to drought stress for 10, 20, and 30 days (D10, D20, D30 for desi; K10, K20, K30 for kabuli) along with control (DC and KC). Relative water content (RWC), Electrolyte leakage index (ELI), and Malondialdehyde (MDA) content under drought stress condition. **b** Chickpea plants under cold stress exposure with details of Thank you for the feedback. We will carefully review and revise the references section to ensure full consistency

Both desi and kabuli chickpea cultivars revealed a significant reduction in relative water content as drought stress increases. In desi chickpea, the RWC of control plants was 83.56% and reported to be decreased to 75.63% at 10% (w/v) PEG, 60.71% at 20% (w/v) PEG, and remained consistent at 60.82% under 30% (w/v) PEG-induced drought conditions (Fig. 9A). In kabuli chickpea displayed a more pronounced decline in RWC. Starting from a control RWC of 87.41%, it dropped to 72.75% at 10% (w/v) PEG, 57.75% at 20% (w/v) PEG, and severely plummeted to 41.17% at 30% (w/v) PEG.

Electrolyte leakage (ELI), is documented as indicator of cellular membrane damage resulting from stress, progressively increased in both chickpea cultivars as the severity of PEG-induced drought stress intensified. In desi chickpea, the ELI was 44.70% in control plant with any stress. Under drought stress, ELI showed a relatively increase, rising to 45.54% (10% (w/v) PEG) to 47.97% (30% (w/v) PEG) suggesting that desi cultivars maintained comparatively stable membrane integrity even under increasing drought conditions. Under cold stress ELI was reported 61.5% in desi chickpea. Conversely, kabuli chickpea exhibited a more pronounced increase in ELI with 49.44%, ELI in controlled plants increases during 54.10% (10% (w/v) PEG) to 66.37% (30% (w/v) PEG). Additionally, ELI was found under cold stress 71.3% in kabuli chickpea. The substantial increase in ELI indicates that the kabuli cultivar was more susceptible to drought stress, having greater membrane damage compared to the desi cultivar.

Malondialdehyde (MDA) content is reported as bio-marker of lipid peroxidation, significantly increased in both desi and kabuli chickpea (*Cicer arietinum* L.) cultivars in response to increasing drought severity and cold stress, indicating enhanced oxidative stress. In desi chickpea, control plants (DC) exhibited an MDA level of 0.0192 nmol/g. This concentration increased substantially to 0.2054 nmol/g under high drought conditions (30% (w/v) PEG, D30) and 0.1401 nmol/g under cold exposure (D Cold). Similarly, kabuli chickpea control plants (KC) showed an MDA content of 0.0430 nmol/g. This increased to 0.2119 nmol/g when subjected to 30% (w/v) PEG-induced drought (K30) and 0.1162 nmol/g was reported under cold stress (Cold). These data conclusively demonstrate a progressive increase in lipid peroxidation and confirm the induction of enhanced oxidative stress in both chickpea cultivars in response to both drought and cold environmental stress.

Differential expression of chickpea *SBT* genes demonstrate kabuli *SBT* genes with high expression response to drought and cold stress compared to desi

To investigate the expression patterns of *subtilase* (*SBT*) encoding genes in chickpea leaf tissues under abiotic stress, a subset of randomly selected genes from both desi and kabuli cultivars was analyzed under drought and cold stress conditions. In the desi cultivar, the expression of *CaDSBT8*, *CaDSBT11*, *CaDSBT27*, *CaDSBT36*, *CaDSBT57*, and *CaDSBT59* was examined (Fig. 9 A). Among these,

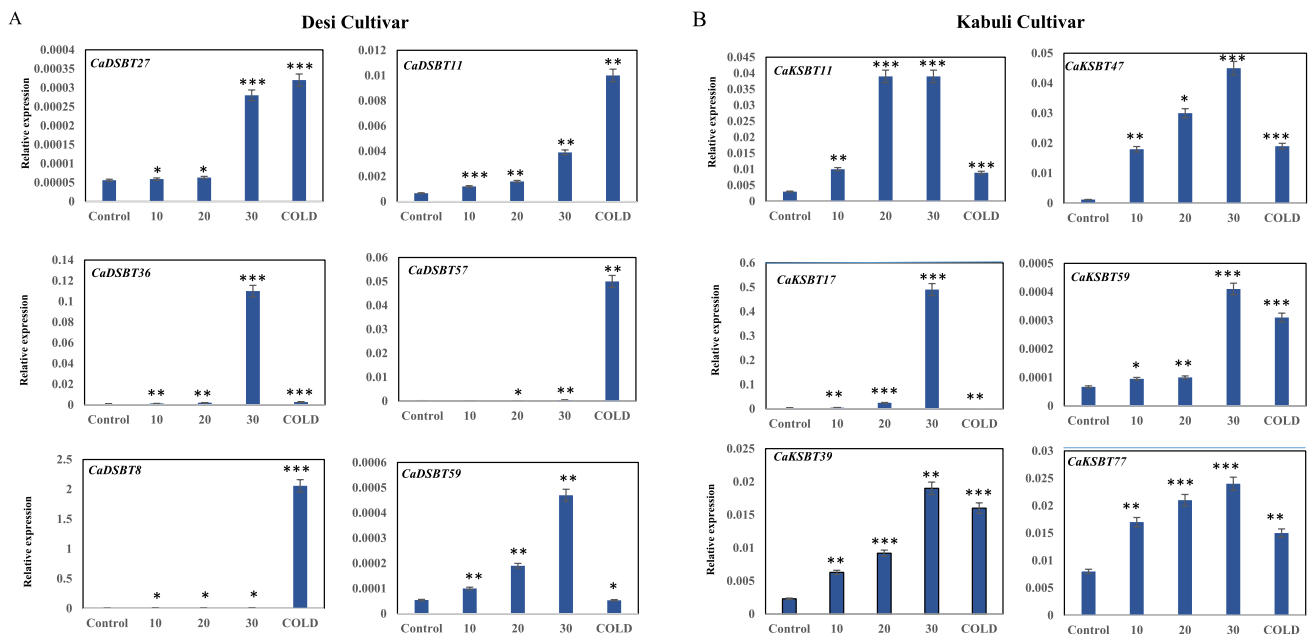


Fig. 9 The expression profiling of selected SBT genes under drought and cold stress using qRT-PCR analysis. **a** Expression analysis of randomly selected *CaDSBTs* in desi cultivar. **b** Expression analysis of randomly selected *CaKSBTs* in kabuli cultivar

CaDSBT11, *CaDSBT27*, and *CaDSBT59* demonstrated significant upregulation under both drought and cold stress. In the kabuli cultivar, the expression profiles of *CaKSBT11*, *CaKSBT17*, *CaKSBT39*, *CaKSBT47*, *CaKSBT59*, and *CaKSBT77* were assessed (Fig. 9 B). Notably, the majority of the *CaKSBT* genes, with the exception of *CaKSBT17*, exhibited significant changes in expression in response to both drought and cold stress. Overall, the kabuli cultivar has been found to more pronounced expression response of its *SBT* genes to both drought and cold stress conditions compared to the desi cultivar, suggesting potentially distinct roles for *SBTs* in stress tolerance between the two chickpea varieties.

Discussion

Plant subtilases (SBTs), a diverse family of serine proteases, are vital for plant growth, development, and a variety of physiological processes (Xu et al. 2019). They are also deeply involved in mediating plant responses to abiotic stresses, such as drought, cold, light, heat and salinity (Wang et al. 2023). This is because abiotic stress can significantly alter the plant proteome, affecting protein abundance, localization, and function (Kosová et al. 2018). While SBT functions have been studied in several legumes, including *Glycine max*, *Vigna radiata*, and *Arachis hypogaea* (Budic et al. 2013; He et al. 2024; Singh et al. 2025), a comprehensive genome-wide analysis of the SBT family in chickpea (*Cicer arietinum*) has been lacking. The present

study addresses this gap by identifying and characterizing the chickpea *SBT* gene family to understand their potential roles in abiotic stress responses.

We identified 69 *CaDSBT* genes in desi chickpea and 82 *CaKSBT* genes in kabuli chickpea, suggesting a greater diversity in the kabuli cultivar. This difference may reflect a combination of domestication-driven selection and adaptation to broader stress environments. These genes were categorized into six groups based on conserved domains and phylogenetic analysis, with most clustering in Group 1. This clustering, along with the dominance of tandem duplication events, suggests that this mechanism has been the primary driver of SBT family expansion in chickpea, which aligns with findings in other legumes (He et al. 2024; Singh et al. 2025; Li et al. 2024). Physicochemical analysis showed that most chickpea SBT proteins are approximately 80 kDa, basic, and hydrophilic, consistent with findings in other species like *Glycine max* and *Arachis hypogaea* (He et al. 2024; Wang et al. 2023; Li et al. 2024). The basic and hydrophilic and stable nature of SBT proteins allows them to function effectively under the harsh and dynamic conditions of the chickpea plant's stress response. Their broad distribution across various cellular compartments, including the chloroplast, cytoplasm, and mitochondria, suggests diverse functional roles critical for an effective stress response (Singh et al. 2025; Dai et al. 2023).

Structural analysis revealed a notable proportion of intronless genes (19 in desi and 15 in kabuli), predominantly in Group 1. These intronless genes could serve as valuable molecular markers for breeding programs focused

on stress tolerance. Differences in protein domains between chickpea and *Arabidopsis* subtilases (SBTs) was observed due to evolutionary divergence and functional specialization. While the core catalytic domains are highly conserved across species, non-catalytic domains evolve more rapidly in response to specific environmental pressures and organismal needs. Furthermore, Gene Ontology (GO) and *cis*-regulatory element (CRE) analyses highlighted the conserved function of chickpea SBTs in key pathways. Similar to *Arachis hypogaea* and *Triticum aestivum* (Li et al. 2024; Zhao et al. 2023), chickpea SBTs are involved in serine-type peptidase activities, indicating a conserved role in protein degradation for maintaining cellular homeostasis and stress adaptation. The presence of specific motifs (e.g. catalytic triads, PA domains) in CaDSBT59/CaKSBT47/CaKSBT77 also highlighted their role in the stress response. The presence of CREs related to stress and phytohormone signaling, particularly methyl jasmonate (MeJA) and abscisic acid (ABA), underscores their regulatory role in these pathways (Hou et al. 2023).

Gene Ontology (GO) enrichment analysis of subtilases (SBTs) in other crops shows their crucial role in stress response via catalytic activities like hydrolase and peptidase function. Specifically, SBTs in peanut (*Arachis hypogaea*) and wheat (*Triticum aestivum*) are involved in serine-type peptidase activities, controlling protein breakdown (Li et al. 2024; Zhao et al. 2023). Our analysis confirms a conserved functional role for chickpea SBTs, which also exhibit serine endopeptidase activity. This conserved function underscores the importance of these enzymes in maintaining cellular homeostasis and helping plants adapt to stress. We also identified cultivar-specific motifs that can be linked to the unique traits of each chickpea variety, providing potential targets for breeding. Analysis of *cis*-regulatory elements (CREs) in chickpea SBT genes revealed associations with plant growth, development, stress responses, and phytohormone signaling. Consistent with findings in other species like maize (*Zea mays*), our results show a high number of CREs related to light, methyl jasmonate (MeJA), and abscisic acid (ABA) signaling (Hou et al. 2023; Li et al. 2024; Jin et al. 2021; Zhao et al. 2023). This suggests chickpea SBTs are regulated by a range of internal and external signals, allowing them to coordinate complex responses to environmental changes.

Our results confirm that chickpea SBTs are involved in stress responses. While drought-tolerant chickpea genotypes typically maintain high relative water content (RWC) (Çevik et al. 2019), we observed a decline in RWC under both drought and cold stress, which was sharper in kabuli cultivars. Both cultivars also showed increased electrolyte leakage (EL) and malondialdehyde (MDA), indicating oxidative stress-induced membrane damage.

In-fact, unregulated SBTs (e.g. *CaDSBT59*) do not directly degrade MDA, but their activity is mechanistically linked to it. Both are indicators of oxidative stress. The link occurs through two primary mechanisms: the direct degradation of proteins damaged by oxidative stress and the mobilization of amino acids to synthesize new stress-response proteins. This aligns with previous studies showing that protection against oxidative stress is a key endurance mechanism in chickpea (Vessal et al. 2020). Previously, Dai et al. (2023) demonstrated that different SBTs play important roles in developmental regulation and stress response mechanisms, including biotic and abiotic stresses. Environmental stresses were found to trigger intricate gene expression cascades that not only confer cellular cytoprotection but also modulate the transcriptional regulation of signal transduction pathways, thereby controlling appropriate physiological and molecular responses (Zhao et al. 2023). Recently, in *Vigna radiata*, *Vradi_SBT-1.9* and *Vradi_SBT-5.6* were demonstrated dramatic transcriptional upregulation (up to 250-fold) and (up to 60-fold) respectively, in response to cold stress, revealing the involvement of *SBT* genes in low-temperature stress responses (Singh et al. 2025). In *Arachis hypogaea*, *AhSBT112*, *AhSBT97*, and *AhSBT98* were significantly upregulated in response to both drought and cold stress while *AhSBT42*, *AhSBT115*, and *AhSBT102* show substantial expression under cold conditions, suggesting their potential role for *SBTs* in cold stress pathways (Li et al. 2024). In present study, expression profiling of *SBT* genes in chickpea leaf tissues under drought and cold stress revealed cultivar-specific and stress-responsive transcriptional dynamics. *CaDSBT27*, *CaDSBT11*, and *CaDSBT59* in desi cultivars and *CaKSBT11*, *CaKSBT39*, *CaKSBT47*, *CaKSBT59*, and *CaKSBT77* in kabuli cultivars were significantly upregulated under both drought and cold stress conditions that is consistent with earlier reports. This upregulation suggests that SBTs play a role in repair processes and cellular protection during stress rather than contribute to stress itself, consistent with findings in other plants (Singh et al. 2025; Li et al. 2024).

The identification of these specific stress-responsive genes and their associated motifs, which can be linked to unique cultivar traits, provides a strong foundation for chickpea breeding. These genes and motifs can be utilized as targets for marker-assisted selection to develop new varieties with enhanced drought and cold tolerance. Additionally, gene editing technologies like CRISPR-Cas can be applied to precisely modulate the expression of these candidate SBT genes to improve stress resilience (Molinari et al. 2021, 2024). This integrated approach can accelerate the development of climate-resilient chickpea varieties, which is crucial for global food security.

Conclusion

This comprehensive study provides an in-depth genome wide identification and expression analysis of Subtilase (SBT) proteins in desi and kabuli chickpea cultivars under their drought and cold stress. Here, 69 CaDSBT and 82 CaKSBT proteins have been identified in desi and kabuli genome respectively. Phylogenetic analysis categorized SBT proteins into six distinct groups, with Group 1 being the largest in both cultivars. Chromosomal mapping showed a largely random distribution across the eight chickpea chromosomes, with chromosome 7 having the highest gene density in both desi (14 genes) and kabuli (20 genes). Gene structure analysis revealed substantial variation in gene length and exon–intron numbers, with a notable presence of intronless genes, particularly in Group 1. Gene duplication analysis, primarily through Ka/Ks ratios, indicated that purifying selection has been the predominant evolutionary force shaping the *SBT* gene family in both cultivars, though one Kabuli gene pair showed evidence of positive selection. Gene Ontology analysis highlighted the involvement of SBT proteins in proteolysis (desi) and broader protein metabolic processes (kabuli), with serine-type endopeptidase activity as a dominant molecular function in both. Cis-regulatory element (CRE) analysis of promoter regions revealed distinct patterns of responsiveness to phytohormones, abiotic stress, and developmental cues. Differential expression analysis confirmed that a subset of *SBT* genes in both cultivars was significantly upregulated under drought and cold stress, with the Kabuli cultivar showing a more pronounced expression response. Hence this study comprehensively characterizes the *SBT* gene family in chickpea, providing valuable insights into their physicochemical properties, genomic organization, evolutionary dynamics, and diverse functional roles. Specifically, these genes could be utilized in modern crop improvement strategies such as genome editing (e.g., CRISPR/Cas9). By precisely editing or manipulating the expression of these key *SBT* genes, we could develop new chickpea varieties with improved drought and cold resistance. These findings contribute significantly to understanding the molecular mechanisms underlying stress tolerance in chickpea and lay a foundation for future genetic improvement strategies. This findings attract future partnerships with biotechnology companies focused on creating climate-resilient crops.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical statement Authors declared that the whole manuscript is strictly complied with ethical standards of Journal of Crop Science and Biotechnology.

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